week2_Research Method_Lecture_Note

Research Method 课堂笔记与翻译

Research Methods: Developing Research Questions

1. Session Introduction and Assessment Overview

Reem: And any questions, feel free to interrupt me. Let's make it interactive, okay?

So today we're going to talk about developing research questions. And this is assessment one of your course.

We have two assessments:

- **Assessment 1**: 60%
- **Assessment 2**: 40%

The first one is entitled "Developing Research Questions".

Before we start the session, I'm just going to tell you in a nutshell what we are going to learn, so that we don't feel like you're lost in the middle of the slides. If you do, feel free to interrupt me.

2. The Foundation of Research: The Question

What we want to know is how to develop **good research questions**.

Last time we said that any research study starts with a question.

What is research? It's answering a question in a systematic and structured way **without bias**. So without having personal opinions. We need to follow a scientific approach to answer questions.

But what is the question? So how do we define questions in the beginning?

3. Sources of Research Questions

The questions that we ask can come from two sources.

Source 1: Practical Problems Either I'm working in a place and I have a **practical problem** that I want to find a solution for, so I'm asking myself how can I solve this problem.

Source 2: Theoretical Gaps (Literature Review) Or it's a **theoretical thing** where I conduct **literature review** and I identify the **gaps** in that literature review, and then I start to answer the questions.

In the coming session, we'll talk more into how to conduct literature reviews. But the idea is, literature reviews is about reading available materials, reading available sources and papers, to identify what are the gaps in that study.

4. Examples of Identifying Gaps

Let's say I'm looking into how, for example, because I come from a digital health background, so my questions might be from the health care background. Still, my background is computer science, so maybe some of the examples would be a little bit different in software engineering.

But let's say I'm looking into the factors that would encourage elderly populations to use mobile health apps. Okay? Let's say I'm looking into that.

So I read the paper. In that paper, there would always be a section related to **limitations**. So what are the limitations of the study? Maybe they say:

- Okay, we didn't take this into consideration.
- Or we studied people in New Zealand only, but not in the wider region of the Oceania, for example.
- Or let's say we considered people between the ages of 65 and 70, not older or younger.

There must be a limitation. Nobody does a study that's completely perfect. Nobody.

So these are **gaps** that you can identify, and then you say, OK, I'm going to extend this research by doing my own research in this space.

5. Practical Problem Example and Course Context

All I told you, it could be another thing, is that you're working in a software development company, and they have a problem. You want to solve this problem. So you're doing research into how to answer this question.

And this is what you're going to be doing in your **capstone projects**, in your final trimester.

So your final trimester, this trimester is a sort of foundational term because people are coming from different backgrounds with different experiences.

研究方法：制定研究问题

1. 课程介绍与考核概述

Reem：有任何问题，请随时打断我。让我们互动起来，好吗?

那么，今天我们将讨论如何制定研究问题。这是你们课程的第一个考核任务。

我们有两项考核：

- **考核一**：占总成绩 60%
- **考核二**：占总成绩 40%

第一项考核的标题是“制定研究问题”。

在开始本次课程之前，我先简要介绍一下我们将要学习的内容，这样大家就不会在幻灯片中感到迷失。如果确实感到困惑，请随时打断我。

2. 研究的基础：问题

我们想要了解的是如何制定**好的研究问题**。

上次我们说过，任何研究都始于一个问题。

什么是研究? 研究是以系统化、结构化的方式，**不带偏见地**回答一个问题。也就是说，不掺杂个人观点。我们需要遵循科学的方法来回答问题。

但问题是什么？那么，我们最初如何定义问题呢?

3. 研究问题的来源

我们提出的问题可以来自两个来源。

来源一：实际问题 可能是我在一个地方工作，遇到了一个**实际问题**，我想找到解决方案，于是我问自己：我该如何解决这个问题。

来源二：理论空白（文献综述） 或者，这是一个**理论性的事情**，我进行**文献综述**，找出该文献综述中的**空白**，然后开始回答这些问题。

在接下来的课程中，我们将更深入地讨论如何进行文献综述。但核心思想是，文献综述就是阅读现有的材料、来源和论文，以确定该研究领域存在哪些空白。

4. 识别研究空白的示例

比如说，我正在研究如何……举个例子，因为我来自数字健康领域，所以我的问题可能来自医疗保健背景。不过，我的背景是计算机科学，所以可能有些例子在软件工程领域会略有不同。

但假设我正在研究鼓励老年人群使用移动健康应用的因素。好吗？假设我正在研究这个。

那么我阅读论文。在那篇论文中，总会有一个与**局限性**相关的部分。那么这项研究的局限性是什么？他们可能会说：

- 好的，我们没有考虑这一点。
- 或者，我们只研究了新西兰的人，但没有研究大洋洲更广泛地区的人。
- 或者，假设我们只考虑了65岁到70岁之间的人，没有考虑更年长或更年轻的人。

一定存在局限性。没有人能做出完全完美的研究。没有人。

所以，这些就是你可以识别的**空白**，然后你说，好的，我打算通过在这个领域进行自己的研究来扩展这项研究。

5. 实际问题示例与课程背景

我刚才提到的，也可能是另一种情况，就是你在一家软件开发公司工作，他们遇到了一个问题。你想解决这个问题。所以你进行研究，以找出如何回答这个问题。

而这正是你们在最后一个学期，在你们的**顶点项目**中将要做的。

所以，你们的最后一个学期，而这个学期更像是一个基础学期，因为大家来自不同的背景，拥有不同的经验。

5. Practical Problem Example and Course Context

I just mentioned that it could also be another scenario where you are working in a software development company and they have a problem. You want to solve this problem. So you conduct research to figure out how to answer this question.

And this is precisely what you will be doing in your final trimester, in your **capstone projects**.

So, your final trimester, and this trimester is more of a foundational term because people are coming from different backgrounds with different experiences.

This is about bringing everyone together to a foundational level.

The next trimester will be more focused on software engineering courses.

And then the last one will involve applying research methods and the courses we learn in software engineering to solve a software-related problem.

6. Session Overview and Learning Objectives

So, in a nutshell, to give you exactly what we're going to do in the coming two hours, we want to learn how to develop research methods.

It starts by reading what's available in the literature, meaning a number of papers.

We need to understand:

- How a paper is designed.
- How we can identify gaps.
- How we build good research questions.
- What makes a research question good or bad?

So this is exactly what we are doing.

We will talk about:

1. The importance of well-formulated research questions.
2. The role of literature review in research question development.

And then, although we will talk about literature reviews in more detail later, this time we are just going to give you an idea about what types of literature reviews exist.

A literature review means reading papers that are relevant to the topic you are trying to research.

So, how do we know that the reading is credible?

- How do I know that this paper is a good paper?
- How do I rely on that?

I cannot go online and find blogs and use them as supporting sources.

This is what we mean by **evaluated scholarly sources**.

If you cite anything, if you write a literature review, it means you are writing about what is already known in that field.

Which means you are using papers that have already been published.

I have to make sure that the papers I am using are credible.

So we are interested in what makes them credible.

Then we will talk about **academic writing**.

So what does it mean to write academically?

I think I mentioned it online. The whole idea is that the way I speak to you is different from when I am writing academically.

When you read research papers, you will think about this. For those of you who downloaded the paper and went through it, you will figure out that the tone is completely different. It is not casual. It is very evidence-based, which means it is supported by references.

When you support your argument with references, it means you are talking scientifically. Otherwise, it would be just chatting or something.

And then, all of these things are related to developing research questions.

So, take a few seconds to go through these learning objectives so that we can all be on the same page.

And note that the slides are available on the Blackboard.

> 👩‍🏫 **Reem**: Just to make sure we're all on the same page. All good? *(Pause)* 👩‍🏫 **Reem**: All good.

So this is how it goes.

We review the literature, meaning we read everything or as much as we can about the topic th...

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5. 实际问题示例与课程背景

我刚才提到，也可能是另一种情况，就是你在一个软件开发公司工作，他们遇到了一个问题。你想解决这个问题。所以你进行研究，以找出如何回答这个问题。

而这正是你们在最后一个学期，在你们的**顶点项目**中将要做的。

所以，你们的最后一个学期，而这个学期更像是一个基础学期，因为大家来自不同的背景，拥有不同的经验。

这是为了让每个人都能达到一个共同的基础水平。

下一个学期将更侧重于软件工程课程。

然后最后一个学期将涉及应用研究方法以及我们在软件工程中学到的课程来解决一个软件相关的问题。

6. 课程概述与学习目标

所以，简而言之，为了确切地告诉你们接下来两小时我们要做什么，我们想学习如何发展研究方法。

它始于阅读文献中已有的内容，也就是阅读大量论文。

我们需要理解：

- 一篇论文是如何设计的。
- 我们如何识别研究空白。
- 我们如何构建好的研究问题。
- 是什么让一个研究问题好或坏？

所以这正是我们要做的。

我们将讨论：

1. 精心构建的研究问题的重要性。
2. 文献综述在研究问题发展中的作用。

然后，虽然我们稍后会更详细地讨论文献综述，但这次我们只是让你们了解一下存在哪些类型的文献综述。

文献综述意味着阅读与你试图研究的主题相关的论文。

那么，我们如何知道所读的内容是可信的？

- 我怎么知道这篇论文是好论文？
- 我如何依赖它？

我不能上网找博客并把它们用作支持来源。

这就是我们所说的**经过评估的学术来源**。

如果你引用任何东西，如果你写一篇文献综述，这意味着你是在撰写该领域已知的内容。

这意味着你正在使用已经发表的论文。

我必须确保我使用的论文是可信的。

所以我们感兴趣的是什么使它们可信。

然后我们将讨论**学术写作**。

那么，学术写作意味着什么？

我想我在网上提到过。核心思想是，我与你交谈的方式与我进行学术写作时是不同的。

当你阅读研究论文时，你会想到这一点。对于那些下载了论文并阅读过的同学，你会发现其语气完全不同。它不是随意的。它非常注重证据，这意味着它由参考文献支持。

当你用参考文献支持你的论点时，意味着你在进行科学的论述。否则，就只是闲聊之类的。

然后，所有这些都与发展研究问题有关。

所以，花几秒钟浏览一下这些学习目标，以便我们都能保持一致。

请注意，幻灯片可在 Blackboard 上找到。

> 👩‍🏫 **Reem**: 只是为了确保我们都在同一页上。都清楚了吗？ *(停顿)* 👩‍🏫 **Reem**: 都清楚了。

所以过程是这样的。

我们进行文献综述，意味着我们阅读关于该主题的所有或尽可能多的内容...

1. Absolute Fidelity (English)

Reem: So take a few seconds to go through these learning objectives so that we can all be on the same page. And note that the slides are available on the Blackboard.

> 👩‍🏫 **Reem**: Just to make sure we're all on the same page. All good? *(Pause)* 👩‍🏫 **Reem**: All good.

So this is how it goes.

We review the literature, meaning we read everything or as much as we can about the topic that we're going to research.

We identify gaps in the papers that we read. So what's missing?

And that's why when we say choose reliable papers, we mean they need to be up to date. Because if you read a paper that's written in 1980, since that time, so many things have changed. So you cannot really rely on that.

So many people have been working on these things, and the results of that paper must not even be relevant. Like even things that I used to cite in 2023 are not relevant with the type of... So that rate of development and the rate of advancement is really going fast, and that's why you need to cite papers that are in the past three years.

Up to five years is fine, but in the assessment we are going to ask you to read papers that are only in the past three years.

We read the paper, we see what are the limitations, what are the things that they haven't done, or what are the things that I can add to, or what are the things that I can try to find out and expand on.

Or the other way would be finding a practical problem in my workspace.

Based on that, I formulate... based on these gaps I formulate the research question. So when I ask you, I do the literature, I decide the gap and develop the research question. Whether it's this term or in the next term, I said remember you always need to say: "This research question is in this gap." Okay?

Systematically, so these are the tasks that I found in the research that require more research, and here I found two to three questions, and these are the two to three questions that I'm going to answer.

So you have to identify what gap is this research question trying to answer.

Then you go and you design your research, meaning then you think about how you're going to collect the data, how you're going to analyze it to answer these questions.

In this session, we'll stop here, okay? I'll keep showing you this diagram just to make sure we're all on the same thing.

So a literature review, you can find the discussion, the definition here, but like who can define literature review in their own words, even casually? What does it mean to conduct a literature review?

> 🙋‍♂️ **Speaker 2**: Thank you.

Just like the example that I gave you when you were trying to make a decision of coming to New Zealand to study a master's degree, right? You did ask questions. You did talk to people about their experiences. You did read things, right?

In a way, this is literature review, but not in a structured academic way. But this is like collecting information about what's already known.

In literature review, you don't come up with results. You collect what's already written and published.

Why published? Because if it's not published, then it's not scientific. It becomes just like opinions, like blogs, social media posts. These are nothing that I can cite and reference to supporting my...

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5. 中文全文翻译

Reem：所以花几秒钟浏览一下这些学习目标，以便我们都能保持一致。请注意，幻灯片可在 Blackboard 上找到。

> 👩‍🏫 **Reem**：只是为了确保我们都在同一页上。都清楚了吗？*(停顿)* 👩‍🏫 **Reem**：都清楚了。

过程是这样的。

我们进行文献综述，意味着我们阅读关于将要研究主题的所有或尽可能多的内容。

我们识别所读论文中的**空白**。那么，缺少了什么？

这就是为什么当我们说选择可靠的论文时，我们指的是它们需要是最新的。因为如果你读一篇写于1980年的论文，从那时起，很多事情都发生了变化。所以你无法真正依赖它。

很多人一直在研究这些问题，那篇论文的结果甚至可能都不再相关了。就像我在2023年引用的一些东西，与现在的类型相比也不相关了……所以这种发展和进步的速度真的非常快，这就是为什么你需要引用过去三年内的论文。

最多五年也可以，但在评估中，我们会要求你只阅读过去三年内的论文。

我们阅读论文，我们看它的局限性是什么，有哪些事情他们还没有做，或者有哪些事情我可以补充，或者有哪些事情我可以尝试去发现和扩展。

另一种方式是在我的工作场所中找到一个实际问题。

基于此，我提出……基于这些空白，我提出研究问题。所以当我要求你们时，我进行文献综述，我确定空白并制定研究问题。无论是这个学期还是下学期，我说过，你们必须始终说明："这个研究问题是为了填补这个空白。"明白吗？

系统性地，这些是我在研究中发现的、需要更多研究的任务，在这里我找到了两到三个问题，而这些就是我要回答的两到三个问题。

所以你必须确定这个研究问题试图回答什么空白。

然后你继续设计你的研究，这意味着你要思考你将如何收集数据，你将如何分析数据来回答这些问题。

这节课我们就讲到这里，好吗？我会继续展示这个图表，以确保我们都理解同一件事。

那么，关于文献综述，你可以在这里找到讨论和定义，但是，谁能用自己的话，哪怕是随意地，定义一下文献综述？进行文献综述意味着什么？

> 🙋‍♂️ **Speaker 2**：谢谢。

就像我之前给你们举的例子，当你们试图决定来新西兰攻读硕士学位时，对吧？你们确实问了问题。你们确实和人们谈论了他们的经历。你们确实阅读了资料，对吧？

在某种程度上，这就是文献综述，但不是以结构化的学术方式。但这就像是收集关于已知事物的信息。

在文献综述中，你不会得出结果。你收集已经撰写和发表的内容。

为什么是发表的？因为如果没有发表，那它就不是科学的。它就变成了像博客、社交媒体帖子一样的观点。这些是我无法引用和参考来支持我的……

1. Absolute Fidelity (English)

But this is like collecting information about what's already known.

In a literature review, you don't come up with results. You collect what's already written and published.

Why published? Because if it's not published, then it's not scientific. It becomes just like opinions, like blogs, social media posts. These are nothing that I can cite and reference to support my argument.

So, a literature review is a critical and comprehensive analysis of existing scholarly work related to a specific topic or research question.

If I'm looking at the negative impacts of social media, this is the topic. I'm going to try to cover the things related to that.

When I look at the research question, I think I'm going to be developed. When I read the literature, I'll start to figure out:

- What's known.
- What's not fully covered.
- What kinds of answers exist.
- Is there a study that answers my question?

When you do a master's, it's just the start of an easier state. When you do your PhD, if you're interested in that, then your life becomes focused on finding these services.

It's very good to synthesize existing knowledge. Not in human knowledge. Revealing inconsistencies.

For example:

- Let's say I found a study that says social media is fabulous. There's no problem with that in the upbringing of kids.
- And another one talks about all the negative things of social media in the quality of life that kids experience.

These are inconsistencies. Then maybe I can come up with a research question that tries to answer: why are there these inconsistencies?

Underexplored areas. Let's say there's not much information about the experiences of specific ethnicities living in New Zealand. For example, I've done my research, but the studies are not really focused on that. So this is an underexplored area. This is a reason for me to research, to develop a question that I want to answer.

Outdated findings. So there's a topic that you like, you've collected information about it, but it's like really old. Or it has been conducted like 10 years ago or 15 years ago. So this is an invitation for you to update the findings. You can support the justification for new inquiry.

When you write a research proposal or a paper, a publication, there's always, before you start talking about the problem that you want to research, you have to do the literature review.

What does it do? It puts the reader in the context of the thing. We're going from general to specific. So it supports the justification for doing inquiry.

How does that so? Usually in papers, when we write papers, we start with the abstract. We write it the last thing because it's like 250 words telling you what this publication is about.

So when I ask you to do a literature review, don't think that I'm expecting you to read like 10 papers fully from beginning to end. No.

What we're expecting is you just go through the abstract.

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5. 中文全文翻译

但这就像是收集关于已知事物的信息。

在文献综述中，你不会得出结果。你收集已经撰写和发表的内容。

为什么是发表的？因为如果没有发表，那它就不是科学的。它就变成了像博客、社交媒体帖子一样的观点。这些是我无法引用和参考来支持我的论点的东西。

因此，文献综述是对与特定主题或研究问题相关的现有学术成果进行批判性和全面的分析。

如果我在研究社交媒体的负面影响，这就是主题。我将尝试涵盖与之相关的内容。

当我审视研究问题时，我认为我将得到发展。当我阅读文献时，我将开始弄清楚：

- 已知的是什么。
- 尚未被充分涵盖的是什么。
- 存在哪些类型的答案。
- 是否有研究能回答我的问题？

当你攻读硕士学位时，这只是一个更轻松状态的开始。当你攻读博士学位时，如果你对此感兴趣，那么你的生活就会专注于寻找这些服务。

综合现有知识是非常好的。不是指人类知识。**揭示不一致性**。

例如：

- 假设我发现一项研究说社交媒体很棒。在孩子的成长过程中没有任何问题。
- 而另一项研究则讨论了社交媒体对孩子在某个年龄段所经历的生活质量的所有负面影响。

这些是不一致之处。那么也许我可以提出一个研究问题，试图回答：为什么存在这些不一致?

未充分探索的领域。假设关于特定族裔在新西兰生活的经历信息不多。例如，我已经做了研究，但这些研究并没有真正聚焦于此。所以这是一个未充分探索的领域。这是我进行研究、提出我想要回答的问题的一个理由。

过时的发现。所以，有一个你喜欢的主题，你收集了相关信息，但它非常陈旧。或者它是在大约10年或15年前进行的。所以这是邀请你更新这些发现。你可以为新的探究提供理由。

当你撰写研究计划书或论文、出版物时，在你开始谈论你想要研究的问题之前，总是必须进行文献综述。

它有什么作用?它将读者置于该事物的背景中。我们从一般到具体。因此，它为进行探究提供了理由。

这是如何做到的?通常在论文中，当我们撰写论文时，我们从摘要开始。我们最后写它，因为它大约250个词，告诉你这篇出版物是关于什么的。

所以，当我要求你们做文献综述时，不要以为我期望你们从头到尾完整地阅读10篇论文。不是的。

我们所期望的是你们只需浏览摘要。

How to Conduct a Literature Review

Reem: Usually in papers, when we write papers, we start with the abstract. We write it the last thing because it's like 250 words telling you what this publication is about.

So when I ask you to do a literature review, don't think that I'm expecting you to read like 10 papers fully from beginning to end. No.

What we're expecting is you just go through the abstract. It's 200 words telling you what this study is about.

You need to find out:

- What are the findings?

- How did they do it?
- In a nutshell, what is this about?

If you feel like you need to know more, if it's relevant to your question, then you can go and read about it. But you scan things, okay? So yeah, so you do that.

Then you do a literature review. At the very end of the literature review section, you say, however this is not done, you identify the limitations. And then you say, this study answers these research gaps.

So the literature review supports why you're building, why you're doing this study.

The Purpose: Identifying Research Gaps

As existing studies are reviewed, the more you read questions or papers, the more you start questioning things, and questions come to your mind, and then you start coming up with the question.

You look for:

- Points of debate.
- Aspects that remain unexamined.

So these gaps represent opportunities to contribute new insights to the field. If what you're doing is already there and already justified and well explored, you're not doing something significant.

The Importance of Novelty and Scientific Verification

So in the proposal, in any proposal, you want to do a study, you need to write a research proposal.

Why? Because studies require a lot of funding. It requires:

- A team of researchers.
- Salaries.
- Tools.
- Travel expenses.
- A lot of things.

So you need to be adding something new to knowledge. The idea is that you're adding to the existing value of knowledge. Then what you have come up with gets added to the database, and somebody else might cite your work in the future.

Reem: Once it's scientifically verified.

> 🙋‍♂️ **Student**: And how do we verify scientifically?
>
> 🧑‍🏫 **Reem**: By a process called **peer review**.

The Peer Review Process

In peer review, what happens? You get experts from the field.

- If it's software engineering, you get people who have been lecturing or working in software engineering for quite a long time.
- If it's in medicine, you get specialists.

This work has to be evaluated by an expert in the material.

What are they looking at? They're looking at flaws. They're looking at what you could have done wrong.

Once they figure out that, okay, this is a robust study, not so many mistakes, not major mistakes, then they approve it. You publish it in a journal, a reputable journal, and it becomes something that any of you can use to defend your arguments in the future.

Recap and Introduction to Review Types

Reem: Right. So, before we talk about that, remember the diagram. I said we start with the literature review, we identify the gaps.

So what are the types of literature review that you can conduct? Depending on the purpose of your study, there are different...

中文全文翻译

如何进行文献综述

Reem：通常在论文中，当我们撰写论文时，我们从摘要开始。我们最后写它，因为它大约250个词，告诉你这篇出版物是关于什么的。

所以，当我要求你们做文献综述时，不要以为我期望你们从头到尾完整地阅读10篇论文。不是的。

我们所期望的是你们只需浏览摘要。它大约200个词，告诉你这项研究是关于什么的。

你需要找出：

- 研究结果是什么？
- 他们是如何做的？
- 简而言之，这是关于什么的？

如果你觉得需要了解更多，如果它与你的问题相关，那么你可以去阅读它。但你们是进行扫描式阅读，明白吗？所以，是的，你们就这样做。

然后你们进行文献综述。在文献综述部分的最后，你们要指出，然而这方面尚未完成，你们要识别出局限性。然后你们说，本研究旨在填补这些研究空白。

因此，文献综述支撑了你为什么要构建、为什么要进行这项研究。

目的：识别研究空白

随着对现有研究的回顾，你阅读的问题或论文越多，你就越开始质疑事物，问题就会浮现在你的脑海中，然后你开始提出你的问题。

你需要寻找：

- 争论点。
- 尚未被检验的方面。

因此，这些空白代表了为该领域贡献新见解的机会。如果你正在做的事情已经存在，并且已经被充分论证和探索，那么你做的就不是什么重要的事情。

新颖性与科学验证的重要性

所以，在提案中，在任何提案中，如果你想做一项研究，你需要撰写一份研究计划书。

为什么？因为研究需要大量资金。它需要：

- 一个研究团队。
- 薪资。
- 工具。
- 差旅费用。
- 很多东西。

所以你需要为知识增添一些新的东西。其理念是你在为现有的知识价值做加法。然后，你所提出的成果会被添加到数据库中，未来可能会有其他人引用你的工作。

Reem：一旦它经过科学验证。

> 🙋‍♂️ **学生**：我们如何进行科学验证？
>
> 👨‍🏫 **Reem**：通过一个叫做**同行评审**的过程。

同行评审过程

在同行评审中，会发生什么？你会得到该领域的专家。

- 如果是软件工程，你会得到那些在软件工程领域教学或工作了很长时间的人。
- 如果是医学，你会得到专家。

这项工作必须由该领域的专家进行评估。

他们看什么？他们寻找缺陷。他们寻找你可能做错的地方。

一旦他们认定，好吧，这是一项严谨的研究，错误不多，没有重大错误，那么他们就会批准。你将其发表在期刊上，一个有声望的期刊上，它就成为你们任何人未来可以用来支持自己论点的东西。

回顾与综述类型介绍

Reem：对。所以，在我们讨论那个之前，记住那个图表。我说过我们从文献综述开始，我们识别空白。

那么，你可以进行哪些类型的文献综述呢？根据你研究的目的，有不同的...

Types of Literature Reviews

Reem: So, what are the types of literature review that you can conduct? Depending on the purpose of your study, there are different types.

We can have:

- A **narrative review**.
- A **systematic review**.
- A **scoping review**.
- A **meta-analysis**.

> 🧑‍🏫 **Lecturer**: In all sessions of research methods, don't think that these are the only examples available. We only give you the most common ones. We can always delve and find other things.

---

1. Narrative Review

A **narrative review** offers a broad overview and summary of a topic.

> 🧑‍🏫 **Lecturer**: Okay? So why am I saying that? I'm comparing these two when I do a systematic one. If I just look up papers... I say I'm looking at papers between 2024 and 2026 in a certain area. These are the keywords I use to retrieve the papers from the databases and so on. This is a general idea.

2. Systematic Review

If I go in a structured, systematic way, I call it a **systematic review**. It usually follows something that we call **PRISMA**.

> 🧑‍🏫 **Lecturer**: Who knows what PRISMA is? Anyone knows what the PRISMA is? 🙋‍♂️ **Student**: (No response indicated) 🧑‍🏫 **Lecturer**: No worries. But it has specific ways and steps of following. So you eliminate this first. It's a set of steps that we will talk about.

In general:

- A **narrative review** offers a broad overview or a summary of what's known about a topic.
- A **systematic review** answers a specific, focused research question.

3. Scoping Review

A **scoping review** maps the breadth of literature on a broad topic for identifying gaps.

> 🧑‍🏫 **Lecturer**: Let's say I want to learn about e-health technologies. So, I see information communication technologies that are used for delivering remote healthcare. I can do a scoping review to understand what is already known and what areas are not covered. What areas are well researched and what areas are not.

4. Meta-Analysis

A **meta-analysis** combines and statistically analyzes the results from multiple studies.

> 🧑‍🏫 **Lecturer**: If I'm looking at existing studies on a certain topic, and I want to say, okay, 20% think that... 70% think... I'm analyzing the existing studies themselves. Okay? So I'm combining statistics and analysis from multiple quantitative studies. Okay? So I'm looking at the already found results and analyzing them.

Summary and Examples

So, the choice of review depends on the purpose, the scope, and the methodology. We'll see examples of that.

Example 1: Narrative Review

> 🧑‍🏫 **Lecturer**: A paper summarizing decades of research on how stress affects academic performance. Summarizing decades of research. So, broad overviews, summarizing existing literature means it's a **narrative review**.

Example 2: Scoping Review

> 🧑‍🏫 **Lecturer**: I review, I define all studies on digital learning platforms and mapping out key things. I'm looking at existing ones and trying to find out, like analyze their results. This is a **scoping review** because it identifies research gaps. What's covered and what's not?

Example 3: Meta-Analysis

> 🧑‍🏫 **Lecturer**: If I'm looking into a study using statistical techniques to combine results from 12 clinical trials on a vaccine. So there are 12 studies on a vaccine efficacy...

文献综述的类型

Reem：那么，你可以进行哪些类型的文献综述呢？根据你研究的目的，有不同的类型。

我们可以有：

- **叙述性综述**。
- **系统性综述**。
- **范围界定综述**。
- **元分析**。

> 🧑‍🏫 **讲师**：在研究方法的整个课程中，不要认为这些是仅有的例子。我们只给你们最常见的那些。我们总是可以深入挖掘并找到其他类型。

---

1. 叙述性综述

叙述性综述 提供对一个主题的广泛概述和总结。

> 🧑‍🏫 **讲师**：明白吗？那我为什么这么说呢？我是在和系统性综述做比较。如果我仅仅是查找论文……我说我要看某个领域2024年到2026年之间的论文。这些是我用来从数据库等地方检索论文的关键词。这是一个总体的想法。

2. 系统性综述

如果我采用一种结构化的、系统性的方式，我称之为 **系统性综述**。它通常遵循我们称之为 **PRISMA** 的指南。

> 🧑‍🏫 **讲师**：谁知道 PRISMA 是什么？有人知道 PRISMA 是什么吗？ 🙋‍♂️ **学生**：（无回应）
> 🧑‍🏫 **讲师**：没关系。但它有特定的遵循方式和步骤。所以你首先排除这个。这是一套我们将会讨论的步骤。

总的来说：

- **叙述性综述** 提供对一个主题已知内容的广泛概述或总结。
- **系统性综述** 回答一个具体的、聚焦的研究问题。

3. 范围界定综述

范围界定综述 描绘关于一个广泛主题的文献广度，以识别研究空白。

> 🧑‍🏫 **讲师**：假设我想了解电子健康技术。那么，我看到用于提供远程医疗的信息通信技术。我可以做一个范围界定综述，以了解已经知道了什么，哪些领域尚未被覆盖。哪些领域研究充分，哪些领域研究不足。

4. 元分析

元分析 结合并统计分析来自多项研究的结果。

> 🧑‍🏫 **讲师**：如果我在看某个主题的现有研究，并且我想说，好吧，20%的人认为……70%的人认为……我是在分析这些现有研究本身。明白吗？所以，我是在结合和分析来自多个定量研究的统计数据。明白吗？所以，我是在查看已经发现的结果并对其进行分析。

总结与示例

因此，综述类型的选择取决于目的、范围和方法论。我们将会看到这方面的例子。

示例 1：叙述性综述

> 🧑‍🏫 **讲师**：一篇总结了几十年来关于压力如何影响学业表现的研究的论文。总结几十年的研究。所以，广泛的概述，总结现有文献，这意味着它是 **叙述性综述**。

示例 2：范围界定综述

> 🧑‍🏫 **讲师**：我综述，我定义所有关于数字学习平台的研究，并描绘出关键内容。我查看现有的研究，并试图找出，比如分析它们的结果。这是一个 **范围界定综述**，因为它识别研究空白。覆盖了什么？没覆盖什么？

示例 3：元分析

> 🧑‍🏫 **讲师**：如果我正在研究一项使用统计技术来结合12项关于疫苗的临床试验结果的研究。所以有12项关于疫苗效力的研究……

Example 2: Scoping Review

🧑‍🏫 **Lecturer**: I'm looking at existing ones and trying to find out, like analyze their results. This is a **scoping review** because it identifies research gaps. What's covered and what's not?

Example 3: Meta-Analysis

🧑‍🏫 **Lecturer**: If I'm looking into a study using statistical techniques to combine results from 12 clinical trials on a vaccine. So there are 12 studies on a vaccine efficacy, for example. I'm going to see the results of study 1, 2, 3, and 12, and then I'm going to analyze them. This is called a **meta-analysis**. It combines findings from multiple studies.

🧑‍🏫 **Lecturer**: Clear?

Example 4: Systematic Review

🧑‍🏫 **Lecturer**: A review following **PRISMA guidelines**. So if you use the PRISMA guidelines, this is very systematic. There's no flexibility. It's very strict. Then, to evaluate effectiveness of a mindfulness app, this is a comprehensive analysis and evidence-based conclusions. This is a **systematic review**.

Summary of Review Types

🧑‍🏫 **Lecturer**: Okay, to summarize:

- If it's very focused and specific, it's a **systematic review**.
- If it's broad and narrative, it's a **narrative review**.
- If it looks at multiple studies to see what's covered the most, it's a **scoping review**.
- If it statistically combines results from multiple studies, it's a **meta-analysis**.

🧑‍🏫 **Lecturer**: So I'll give you five seconds. Just go through these ones. I always highlight the important keywords. Keep them in your mind, and I'm going to ask you on the next slide to categorize a number of studies based on whether they're... talking with this analysis of autism.

Interactive Exercise: Testing Your Understanding

🧑‍🏫 **Lecturer**: So now it's time for you to test your understanding. These are eight examples. But anyway, so read these ones. If you want to go to the next slide, let me know.

Class Roll Call

👨‍🏫 **Lecturer**: Adiyah? Who's last name is Adiyah? It's B-A-I-H-T-I-A-T-I. 👨‍🏫 **Lecturer**: Bad Paul? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Bimah? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Darling? 👨‍🏫 **Lecturer**: Mingong? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Mingong? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Lazmanisha? 🙋‍♂️ **Student**: I'm here. 👨‍🏫 **Lecturer**: Miral? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Nihiniya? 🙋‍♂️ **Student**: Sorry. 👨‍🏫 **Lecturer**: Did I say it right? 🙋‍♂️ **Student**: Nehemiah. 👨‍🏫 **Lecturer**: It's for him. Okay. 👨‍🏫 **Lecturer**: Penguin? 🙋‍♂️ **Student**: No. 👨‍🏫 **Lecturer**: Ravinda, Mansika, Ravinda Sunar. Roshan, Sandali, President, Sandal, Sarita, Sarita, Sarifa? 👨‍🏫 **Lecturer**: Sean? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Gibran? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Sayid Abdullah? 🙋‍♂️ **Student**: Yes. 👨‍🏫 **Lecturer**: Weefe? 🙋‍♂️ **Student**: Do you... I'm sorry. 👨‍🏫 **Lecturer**: Upside? Wengi? Wenji? Wenji? Wenji? You are you? 🙋‍♂️ **Student**: Wenji? 🙋‍♂️ **Student**: You are Ziyu. Ok. You are saying Chinese. I'm sorry.

例 2：范围界定综述

👨‍🏫 **讲师**：我查看现有的研究，并试图找出，比如分析它们的结果。这是一个 **范围界定综述**，因为它识别研究空白。覆盖了什么？没覆盖什么？

例 3：元分析

👨‍🏫 **讲师**：如果我正在研究一项使用统计技术来结合12项关于疫苗的临床试验结果的研究。所以有12项关于疫苗效力的研究，例如。我将查看研究1、2、3和12的结果，然后分析它们。这被称为 **元分析**。它结合了来自多项研究的发现。

👨‍🏫 **讲师**：清楚了吗？

例 4：系统综述

👨‍🏫 **讲师**：遵循 **PRISMA 指南** 的综述。所以如果你使用 PRISMA 指南，这是非常系统化的。没有灵活性。非常严格。然后，为了评估正念应用程序的有效性，这是一个全面的分析和基于证据的结论。这是一个 **系统综述**。

综述类型总结

👨‍🏫 **讲师**：好的，总结一下：

- 如果它非常聚焦和具体，就是 **系统综述**。

- 如果它是宽泛和叙述性的，就是 **叙述性综述**。
- 如果它查看多项研究以了解覆盖最多的内容，就是 **范围界定综述**。
- 如果它统计地结合了多项研究的结果，就是 **元分析**。

🧑‍🏫 **讲师**：所以我给你们五秒钟。快速浏览一下这些。我总是强调重要的关键词。把它们记在脑子里，我将在下一张幻灯片上要求你们根据它们是否……谈论这个自闭症分析来对一系列研究进行分类。

互动练习：测试你的理解

🧑‍🏫 **讲师**：现在是时候测试你的理解了。这里有八个例子。总之，阅读这些。如果你想看下一张幻灯片，请告诉我。

课堂点名

🧑‍🏫 **讲师**：Adiyah? 谁的姓氏是 Adiyah? 是 B-A-I-H-T-I-A-T-I。 🧑‍🏫 **讲师**：Bad Paul? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Bimah? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Darling? 🧑‍🏫 **讲师**：Mingong? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Mingong? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Lazmanisha? 🙋 **学生**：我在这里。 🧑‍🏫 **讲师**：Miral? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Nihiniya? 🙋 **学生**：抱歉。 🧑‍🏫 **讲师**：我说对了吗? 🙋 **学生**：Nehemiah。 🧑‍🏫 **讲师**：是叫他。好的。 🧑‍🏫 **讲师**：Penguin? 🙋 **学生**：不是。 🧑‍🏫 **讲师**：Ravinda, Mansika, Ravinda Sunar。Roshan, Sandali, President, Sandal, Sarita, Sarita, Sarifa? 🧑‍🏫 **讲师**：Sean? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Gibran? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Sayid Abdullah? 🙋 **学生**：到。 🧑‍🏫 **讲师**：Weefe? 🙋 **学生**：你……抱歉。 🧑‍🏫 **讲师**：Upside? Wengi? Wenji? Wenji? Wenji? 你是你吗? 🙋 **学生**：Wenji? 🙋 **学生**：你是 Ziyu。好的。你在说中文。抱歉。

Research Methods Course: Literature Review Types - Examples & Discussion

Attendance & Course Roll Call

🧑‍🏫 **Lecturer**: Sayid Abdullah? 🙋 **Student**: Yes.

🧑‍🏫 **Lecturer**: Weefe? 🙋 **Student**: Do you... I'm sorry.

🧑‍🏫 **Lecturer**: Upside? Wengi? Wenji? Wenji? Wenji? You are you? 🙋 **Student**: Wenji? 🙋 **Student**: You are Ziyu. Ok. You are saying Chinese. I'm sorry.

🧑‍🏫 **Lecturer**: Thank you.

Absent Students: Wenji is absent, Adia is absent, and Sovji is absent.

Reviewing Literature Review Types: Practical Examples

Example 1: Scoping Review

A researcher wants to explore and summarize various psychological studies.

- The goal is to map a broad topic area over a long time.
- **This is a scoping review.**

Example 2: Systematic Review

A psychologist aims to compare the outcomes.

- **Correct.**
- **Systematic.**

Example 3: Narrative Review

A communication studies scholar examines how social media influences political discourse.

- They integrate information from academic articles, opinion essays, and grey literature to argue how these platforms change public opinion.

> 🙋 **Student**: Net analysis. 👨‍🏫 **Lecturer**: Why net analysis? 👨‍🏫 **Lecturer**: It's a narrative review. But what makes you think it's a narrative one? 🙋 **Student**: I thought because there are academic articles, opinion essays... 👨‍🏫 **Lecturer**: I can see why you said that, but in **meta-analysis**, you're basically statistically analyzing the results of each of the studies, and then you're analytically analyzing them. Okay?

Example 4: Scoping Review

An education researcher is investigating how artificial intelligence is being used in classrooms.

- **What's that?**
- **Scoping. Scoping.**

Example 5: Systematic Review

A health researcher investigates whether mindfulness reduces stress.

- **Why?** Because we have three defined protocols, strict inclusion criteria—very strict.
- Remember when I told you we will learn about the **PRISMA** literature review? You'll be able to use it really well in your assessment, but it's very strict.
- It's like guidelines, steps that you need to follow. This is where we call it a **systematic review**.
- And usually, it's a **PRISMA approach**.

Example 6: Meta-Analysis

A nutrition scientist reviews multiple experimental studies on the effects of a specific diet.

- **What's that?**
- **Meta.** Because we're looking into the results of each one of them and trying to analyze them statistically.

> 👨‍🏫 **Lecturer**: Well done. Right.

Importance of Critical Evaluation

So, evaluating the credibility of the sources is very important.

- This is very important for this course, for other courses, and for your future work.

> 👨‍🏫 **Lecturer**: Okay? I really keep coming back to Trimester 3. I know it's early stages, but keep these in mind because next trimester, you'll be doing so many practical things that you might miss it.

研究方法课程：文献综述类型 - 示例与讨论

考勤与课程点名

> 👨‍🏫 **讲师**：Sayid Abdullah? 🙋 **学生**：到。

> 🧑‍🏫 **讲师**：Weefe? 🙋‍♂️ **学生**：你……抱歉。
>
> 🧑‍🏫 **讲师**：Upside? Wengi? Wenji? Wenji? Wenji? 你是你吗? 🙋‍♂️ **学生**：Wenji? 🙋‍♂️ **学生**：你是 Ziyu。好的。你在说中文。抱歉。
>
> 🧑‍🏫 **讲师**：谢谢。

缺席学生：Wenji 缺席，Adia 缺席，Sovji 缺席。

回顾文献综述类型：实践示例

示例 1：范围界定综述

一名研究人员希望探索和总结各种心理学研究。

- 目标是绘制一个广泛主题领域在长时间内的概况。
- **这是一个范围界定综述。**

示例 2：系统综述

一名心理学家旨在比较不同干预措施的结果。

- **正确。**
- **系统综述。**

示例 3：叙述性综述

一名传播学学者研究社交媒体如何影响政治话语。

- 他们整合了学术文章、观点文章和灰色文献中的信息，以论证这些平台如何改变公众舆论。

> 🙋‍♂️ **学生**：网络分析。🧑‍🏫 **讲师**：为什么是网络分析? 🧑‍🏫 **讲师**：这是一个叙述性综述。但什么让你认为它是叙述性的? 🙋‍♂️ **学生**：我以为是因为有学术文章、观点文章…… 🧑‍🏫 **讲师**：我明白你为什么这么说了，但在**元分析**中，你基本上是在对每项研究的结果进行统计分析，然后进行综合分析。明白吗?

示例 4：范围界定综述

一名教育研究人员正在调查人工智能如何在课堂中使用。

- **这是什么类型?**
- **范围界定综述。范围界定综述。**

示例 5：系统综述

一名健康研究人员调查正念是否能减轻压力。

- **为什么?** 因为我们有三个明确的协议，严格的纳入标准——非常严格。
- 记得我告诉过你们我们将学习 **PRISMA** 文献综述吗？你们将能在评估中很好地使用它，但它非常严格。
- 它就像指南，你需要遵循的步骤。这就是我们称之为**系统综述**的地方。
- 通常，它采用 **PRISMA 方法**。

示例 6：元分析

一名营养科学家回顾了关于特定饮食效果的多个实验研究。

- **这是什么类型?**
- **元分析。** 因为我们要查看每一项研究的结果，并试图对它们进行统计分析。

> 🧑‍🏫 **讲师**：做得好。对的。

批判性评估的重要性

因此，评估来源的可信度非常重要。

- 这对本课程、其他课程以及你未来的工作都非常重要。

> 🧑‍🏫 **讲师**：明白吗？我真的会不断提到第三学期。我知道现在还处于早期阶段，但请记住这些，因为下学期你们会做很多实践性的事情，可能会忽略这一点。

Credible Sources and How to Find Them

Whenever you write an assessment that requires you to support your arguments, the sources have to be credible.

By credible, I mean **peer reviewed**, so reviewed by experts before publication. This is what we find in academic journals.

Using Google Scholar

I want you to go to **Google Scholar**.

You can identify which years you want to search in. For example, from 2022 or 2000 to 2026. You can always identify which papers you wanted to read.

And see these peer-reviewed ones. Whenever you find them in journals, this is a good thing. Because it wouldn't be published unless it's very scrutinized.

Other Research Databases

In this database, the more you go into the research space, the one who gets used to the famous one. So I believe all you guys, it's called the **Web of Science**.

These are the databases that you can search with.

It has been involved in journals with high **impact factors**.

Understanding Journal Quality and Impact Factors

Even the journals are not all good. We have **Q1, Q2, Q3, Q4**.

- **Q1 is the top.**

Usually the Q1 ones have high impact factors, which tells you how many citations have been used in that journal. So how many times it has been cited in research?

The more I use your arguments to support my work, it means I trust you, right?

So it's the average number of citations received by articles in a particular journal of a specific year. It's what it told you.

And what does it tell you? It tells you the **influence** of the journal, the **visibility**, and the **impact**.

🧑‍🏫 **Lecturer**: So if I tell you I've published something in *Sustainability*, for someone in the discipline, oh, that's great. Because we already know, you get used to the good one.

The Importance of Publication Date

Data publication, as I told you. It could be a very good one, but outdated. So what's the point? Maybe so many advances have happened since then.

So ideally **five to 10 years**, except foundational or historical studies.

- I can use a paper to use a definition. For example, I can use a paper to define what a disaster is. And that definition would be earlier than that. That's fine. What definitions were these.
- But when I say 65% of people did that, this requires to be like really recent.

Evaluating a Paper's References

Referring to my citations. One of the things that we look at when we evaluate research papers is that we look at the list of citations that the authors have used to support their arguments.

So in a paper you have the abstract, the literature review or introduction...

可信来源及其查找方法

每当你撰写一份需要你支持自己论点的评估报告时，所使用的来源必须是可信的。

所谓可信，我指的是**经过同行评审的**，即在发表前经过专家评审的。这就是我们在学术期刊中找到的内容。

使用谷歌学术

我希望你们去使用**谷歌学术**。

你可以指定要搜索的年份范围。例如，从2022年或2000年到2026年。你总是可以确定你想阅读哪些论文。

并且查看这些经过同行评审的。每当你在期刊中找到它们，这是一件好事。因为除非经过严格审查，否则它不会被发表。

其他研究数据库

在这个数据库领域，你越深入科研领域，就会习惯使用那个著名的。所以我相信你们都知道，它叫做**Web of Science**。

这些是你可以用来搜索的数据库。

它收录了具有高**影响因子**的期刊。

理解期刊质量与影响因子

即使是期刊，也并非都是好的。我们有**Q1, Q2, Q3, Q4**。

- **Q1是顶级的。**

通常Q1期刊具有高影响因子，它告诉你该期刊中的文章被引用了多少次。也就是说，它在研究中被引用了多少次?

我越多地使用你的论点来支持我的工作，就意味着我信任你，对吗?

所以，它指的是特定年份中，某期刊文章所获得的平均引用次数。这就是它所告诉你的。

它告诉你什么？它告诉你期刊的**影响力**、**可见度**和**影响**。

> 🧑‍🏫 **讲师**：所以，如果我告诉你我在*Sustainability*期刊上发表了一些东西，对于该领域的人来说，哦，那太棒了。因为我们已经知道，你习惯了好的期刊。

出版日期的重要性

正如我告诉你的，数据发表。它可能是一篇非常好的文章，但已经过时了。那有什么意义呢?也许自那以后已经发生了许多进展。

所以理想情况下是**五到十年内**，除了基础性或历史性研究。

- 我可以用一篇论文来使用一个定义。例如，我可以用一篇论文来定义什么是灾难。那个定义可能比这更早。这没问题。这些定义是什么。

- 但是当我说65%的人做了某事时，这需要是非常近期的数据。

评估论文的参考文献

提到我的引用。我们在评估研究论文时关注的事情之一，就是查看作者用来支持其论点的引用列表。

所以在一篇论文中，你有摘要、文献综述或引言...

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所以在一篇论文中，你有：

- 摘要
- 文献综述或引言
- 方法论（即你如何收集和分析数据）
- 研究发现
- 对研究发现的讨论
- 局限性及未来工作

在出版物的最后，你会有一个你在论文中用来支持论点的参考文献列表。

这个列表也告诉我们你研究的质量：

- 你使用了哪些期刊?
- 你是否只使用了会议论文？会议论文通常不如期刊论文严格。
- 你使用的是哪种类型的期刊？等等。

所以，也要查看文献中使用的参考文献列表。

研究方法论是关键

我们将讨论研究方法论，即你如何收集和分析数据。

我从我的KSC导师那里得到的一条建议，对我的职业生涯帮助很大：

- **不要使用大词，不要试图显得迷人、令人印象深刻。**
- 越简单，越能让我（作为审阅你工作的人）相信你知道自己在做什么。
- 使用大词、试图表达宏大的想法，并不能说服该领域的专家。
- 使用**简单、清晰的英语**。

因此，如果方法论清晰，我能轻松理解你如何分析数据，这就是判断其为**可信信息来源**的一个非常重要的标准。

资金与潜在偏见

这是什么意思？

通常，如果有人资助任何研究，你必须（在伦理上）在文末注明，例如“本研究由商业部、奥克兰大学等资助”。

所以，如果我读到一项研究，该研究告诉我“这些对健康没那么坏”。然后我往下看，发现资助方是可口可乐公司。

这确实会引发一个大问题，对吧？这意味着，好吧，这不可信，因为结果朝这个方向发展符合他们的利益。

所以，**资金和潜在偏见**很重要。

研究的可靠性

总的来说，所有这些都使研究建立在**可靠、高质量的证据**之上。

所以，当我要求你们查找论文时，你们应该仔细审视每一篇论文。

所有的研究……我们接下来会讨论，但总的来说，**研究也可能存在偏见。**

你在进行研究时必须审视自己。

> **Reem**: 好吧？

假设我来自……比方说，为了讨论方便，假设我在医疗系统中有过糟糕的经历。或者医生对我不好，等等。

然后我进行了一项关于新西兰人接受医疗保健的看法的研究。

作为一个人，我的经历可能会影响我的分析。

> 🧑‍🏫 **Reem**: 对吧?

这被称为**偏见**。

中文全文翻译

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> 🧑‍🏫 **Reem**: 好吧?

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作为一个人，我的经历可能会影响我的分析。

> 🧑‍🏫 **Reem**: 对吧?

这被称为**偏见**。

Bias in Research

> 🧑‍🏫 **Reem**: And in that situation, we're not denying that it happens. We know that it happens. We're just embracing it. We're being concerned about it.

The reader needs to know that the person conducting the research has relevant experiences.

For example, not necessarily personal, but you could state: "I have been working with Coca-Cola."

This is to give an indication of what kind of **bias** might be present in this research.

- **Bias distorts data collection and analysis.**
- It can be **intentional or unintentional**.
- Sometimes it is intentional.

A Note on Intentional Bias and Cultural Awareness

Let me tell you something about this intentional bias.

If you remember, I told you about **cultural awareness** and the city of Aitaini in the first online session.

- New Zealand's indigenous people are the **Māori people**.
- They were colonized by the British.
- After much conflict, an agreement was reached on how to coexist in New Zealand.

This agreement is based on three foundational principles, often called the **Three P's**:

- Participation
- Protection

- ... (I forgot the last P)

> 🧑‍🏫 **Reem**: Okay, these are the foundations of this concept.

This concept has implications for everything we do. Go and learn about this because if you go to a job interview, they might ask you:

- What you know about these principles.
- How you would apply these concepts in your writing.

It could be a given question.

The Importance of Cultural Awareness in Research

> 🧑‍🏫 **Reem**: Okay?

So, when we do research in a **Māori context**, or when we say we have to be **culturally aware**, it is very important.

The reason is that these communities are very sensitive regarding their data—how it is collected, used, and connected.

This is not because they are difficult. They have had varied experiences with data.

- **Earlier Example:** They were very generous with data, giving any data to researchers.
- **Consequence:** The results were **biased** and did not reflect reality.

Example: Misrepresentation in Health Research

I'll give you an example.

For instance, Māori and Pasifika peoples are known to have high rates of obesity.

A simplistic research conclusion might be: "People need to eat well and exercise."

> 🧑‍🏫 **Reem**: You're not taking into consideration that junk food might be cheaper than healthy food. So you're misrepresenting things. You're presenting your results in a biased

way.

So anyway, this is just something that links to you.

Conclusion: Why We Focus on This

This is why we focus so much on the cultural aspects of Māori research.

It's also why we are going to invite an expert to tell us very important stuff about this.

> 👨‍🏫 **Reem**: So in general, research could be biased, we have to pay attention to that, and...

研究中的偏见

> 👨‍🏫 **Reem**: 在这种情况下，我们并非否认它的发生。我们知道它会发生。我们只是正视它。我们对此表示关切。

读者需要知道进行研究的人具有相关的经验。

例如，不一定是个人经验，但你可以声明："我曾在可口可乐公司工作过。"

这是为了表明这项研究中可能存在哪种**偏见**。

- **偏见会扭曲数据收集和分析。**
- 它可以是**有意的或无意的**。
- 有时它是有意的。

关于有意偏见与文化意识的说明

让我告诉你们一些关于这种有意偏见的事情。

如果你们还记得，我在第一次在线课程中告诉过你们关于**文化意识**和艾泰尼市的事情。

- 新西兰的原住民是**毛利人**。
- 他们曾被英国殖民。
- 经过许多冲突后，达成了关于如何在新西兰共存的协议。

该协议基于三个基本原则，通常被称为**"三个P"**：

- 参与 (Participation)
- 保护 (Protection)
- ... （我忘了最后一个P）

> 🧑‍🏫 **Reem**: 好吧，这些是这个概念的基础。

这个概念对我们所做的一切都有影响。去了解一下这个，因为如果你去参加求职面试，他们可能会问你：

- 你对这些原则了解多少。
- 你将如何在你的写作中应用这些概念。

这可能是一个给定的问题。

研究中文化意识的重要性

> 🧑‍🏫 **Reem**: 好吗?

因此，当我们在**毛利背景**下进行研究，或者当我们说我们必须具备**文化意识**时，这是非常重要的。

原因是这些社区对他们的数据非常敏感——数据如何被收集、使用和关联。

这并不是因为他们难以相处。他们在数据方面有过不同的经历。

- **早期的例子：** 他们过去对数据非常慷慨，向研究人员提供任何数据。
- **后果：** 结果是**有偏见的**，并不能反映现实。

示例：健康研究中的错误表述

我给你们举个例子。

例如，众所周知，毛利人和太平洋岛民肥胖率很高。

一个简单化的研究结论可能是："人们需要吃得好并锻炼身体。"

> 🧑‍🏫 **Reem**: 你没有考虑到垃圾食品可能比健康食品更便宜。所以你是在错误地表述事情。你是在以一种有偏见的方式呈现你的结果。

总之，这只是与你们相关的一点。

结论：我们为何关注这一点

这就是为什么我们如此关注毛利人研究的文化方面。

这也是为什么我们将邀请一位专家来告诉我们关于这方面非常重要的信息。

> 🧑‍🏫 **Reem**: 所以总的来说，研究可能存在偏见，我们必须注意这一点，并且...

1. Absolute Faithfulness (English)

Reem: So you're misrepresenting things. You're presenting your results in a biased way.

So anyway, this is just something that relates to you.

Conclusion: Why We Focus on This

This is why we focus so much on the cultural aspects of Māori research.

This is also why we are going to invite an expert to tell us about this very important topic.

> 🧑‍🏫 **Reem**: So in general, research could be biased, we have to pay attention to that, and there are techniques that we will learn to help us minimize bias and identify it ourselves.

So if a study is biased, it's not credible.

How do we spot bias?

- We look at the methodology.
- We look at the sample sizes. For example, if you're looking at the experiences of women in New Zealand, have we considered everyone, every woman? Or are we only focused on a certain age group, a certain ethnicity, or a certain discipline?
- Some of the times, these factors have to be included.
- Funding disclosures.

When you read the findings and the discussions, have the authors discussed all the findings, or did they just focus on the 10 things they preferred?

Okay, so this is in general. There are things to consider.

Where to Find Papers for Your Assessment

I believe computer science is very relevant to our software engineering.

Key Resources:

- There is a database called **ESCO**. I think the university will give you access to it through the UB network, but you have to be on campus to access it.
- And **O'Reilly** as well.
- These are platforms where you can look up papers.

Google Scholar is like a go-to place any researcher would start with.

Note on Google Scholar:

- In Google Scholar, you might not find full papers. You might only be able to read the abstract, and then you have to pay to unlock the whole paper.
- Sometimes they are free.
- In research nowadays, there is a move away from paid subscriptions. The idea is that knowledge should be available for everyone. So there's a trend towards not having to pay a lot of money to access research.
- Google Scholar has all kinds of papers.

Assessment Task Overview

In the assessment, we are going to ask you to:

1. Choose a paper.
2. Do a literature review.
3. Identify gaps.
4. Build your research questions.

This starts with **choosing a good paper**. So please, please pay attention.

And by the way, the mark is dedicated for us. So you can pick the mark. This is a given. So don't choose a paper... So, yeah.

I'll give you a minute break so that we don't overwhelm you with these ideas.

Okay, today we will come back to talk about identifying the research gap. We'll see you at 4:20.

Thank you.

> 🙋 **Speaker 2**: I can't see the camera.

Reem: I'm the first one. It's a free one.

> 🙋 **Speaker 2**: The Chinese has been 1000 more.

Reem: First, we have to define the time. Face-to-face and the 5% of your time during the week. You should be learning the time. Okay?

Okay, so, again as I said, the gap represents the opportunity for new research to be there.

Now, we usually say, here's the literature table, read it...

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5. 中文全文翻译

Reem: 所以你是在错误地表述事情。你是在以一种有偏见的方式呈现你的结果。

总之，这只是与你们相关的一点。

结论：我们为何关注这一点

这就是为什么我们如此关注毛利人研究的文化方面。

这也是为什么我们将邀请一位专家来告诉我们关于这方面非常重要的信息。

> 🧑‍🏫 **Reem**: 所以总的来说，研究可能存在偏见，我们必须注意这一点，并且我们将学习一些技巧来帮助我们最小化偏见并自行识别它。

所以，如果一项研究存在偏见，它就是不可信的。

我们如何发现偏见?

- 我们审视研究方法论。
- 我们审视样本量。例如，如果你在研究新西兰女性的经历，我们是否考虑了所有人、每一位女性？还是我们只关注某个特定的年龄组、特定的种族或特定的学科?
- 有时，这些因素必须被纳入考虑。
- 资金披露。

当你阅读研究结果和讨论部分时，作者是否讨论了所有的发现，还是他们只关注了他们偏好的那10件事?

好的，总的来说就是这样。有些事情需要考虑。

在哪里为你的评估查找论文

我相信计算机科学与我们的软件工程高度相关。

关键资源:

- 有一个名为 **ESCO** 的数据库。我想大学会通过UB网络给你们访问权限，但你们必须在校园内才能访问。
- 还有 **O'Reilly**。
- 这些是你可以查找论文的平台。

Google Scholar 就像是任何研究者都会首先使用的首选之地。

关于 Google Scholar 的注意事项:

- 在 Google Scholar 中，你可能找不到全文。你可能只能阅读摘要，然后需要付费才能解锁整篇论文。
- 有时它们是免费的。
- 在当今的研究中，有一种远离付费订阅的趋势。理念是知识应该对所有人开放。因此，有一种趋势是不需要支付大量金钱来获取研究。
- Google Scholar 有各种类型的论文。

评估任务概述

在评估中，我们将要求你们：

1. 选择一篇论文。
2. 进行文献综述。
3. 找出研究空白。
4. 构建你的研究问题。

这始于**选择一篇好的论文**。所以请，请大家务必注意。

顺便说一下，分数是给我们预留的。所以你可以选择这个分数。这是给定的。所以不要选择论文... 所以，是的。

我给你们一分钟休息时间，以免这些想法让你们感到压力过大。

好的，今天我们将回来讨论如何识别研究空白。我们4点20分见。

谢谢。

> 🙋‍♂️ **Speaker 2**: 我看不到摄像头。

Reem: 我是第一个。它是免费的。

> 🙋‍♂️ **Speaker 2**: 中文的已经超过1000了。

Reem: 首先，我们必须定义时间。面对面以及你每周5%的时间。你应该在学习时间。好吗?

好的，所以，正如我再次说过的，空白代表了新研究存在的机会。

现在，我们通常会说，这是文献表格，阅读它...

Research Methods - Session Content

1. Course Logistics & Exercise Clarification

Reem: First, we have to define the time. Face to face and the 5% of your time during the week. You should be learning during that time. Okay?

Okay, so, again as I said, the gap represents the opportunity for new research to exist.

Now, we usually say, here's the literature table, read it, identify the limitations or the gap that you can figure out, but we don't really delve into this in depth during the session.

So this is a design exercise. If you do this exercise, you save time for the assessment.

My plan was to have the first talk for the content, and then when we come back from the break we work on it. But I have a lot to cover, so I'm not sure if we're going to have time or not. If we do, that would be great.

It's posted under self-directed learning on Blackboard. I strongly encourage you to go through that exercise.

Regarding working groups, it's not like I'm going to ask you to submit something and it's going to be graded. It's just for you. But it will help you deliver a very good assessment.

Cool? Right.

2. Core Academic: Understanding Critical Analysis

So, when we say **critical analysis**, it doesn't mean some of the other things. It doesn't mean summarizing what you read.

It's about **evaluating** the research by examining its **strengths and weaknesses**.

I can imagine that this might be a little bit challenging for someone who's just started research. But when you critically look at something, you're not just reading it and summarizing it.

You are:

- Looking at identifying the **biases**, the **gap**.
- Comparing it with other studies that you've read.
- Assessing the **quality of the evidence**.

So what are the sources that they used to come up with and write their papers? You are questioning the method.

So think about it as reading something to try and figure out, try and find the fault with it. That's a negative way of looking at it, but practically speaking, it's like that.

You are looking at something to see:

- What's unusual?
- What's the gap?
- What's the comparison?
- What do other studies say in this aspect?

So this is **critical analysis**. We're being critical about something. And it's different than summarizing the findings.

Remember, the instruction may say "critically analyze the papers." OK?

3. Example: Summary vs. Critical Analysis

I'll give you an example. Something that's not critical. And then I'll show you what a critical analysis would look like.

Smith, 2020. So this is how we cite papers. In the self-directed learning, you should know how to do APA referencing. **APA** is the way we use citations in EOE. Okay?

So when you find a paper, there are different ways of analyzing it. Here is an example of a **non-critical summary**:

> "Smith (2020) found that remote work increases productivity by 15%. The study surveyed 500 employees and used statistical analysis. This shows that remote work is beneficial for companies."

This is a summary of the study. There's nothing critical here.

研究方法 - 课程内容

1. 课程事务与练习说明

Reem: 首先，我们必须明确时间。面对面授课以及你每周5%的学习时间。你应该在那段时间里进行学习。明白吗?

好的，所以，正如我再次说过的，空白代表了新研究存在的机会。

通常我们会说，这是文献表格，阅读它，找出你能发现的局限或空白，但我们不会在课程中深入探讨这一点。

所以这是一个设计练习。如果你完成这个练习，你就能为评估节省时间。

我原本的计划是先讲解内容，然后休息回来后我们一起做这个练习。但我有很多内容要讲，所以我不确定我们是否有时间。如果有，那就太好了。

它发布在Blackboard的自主学习部分。我强烈建议你们完成那个练习。

关于小组作业，这并不是说我会要求你们提交什么东西并给它打分。这只是为你们自己准备的。但它将帮助你完成一份非常好的评估作业。

清楚了吗？好的。

2. 核心学术：理解批判性分析

那么，当我们说 **批判性分析** 时，它并不意味着其他一些事情。它并不意味着总结你读到的内容。

它是关于通过审视研究的 **优点和缺点** 来 **评估** 这项研究。

我可以想象，这对于刚开始做研究的人来说可能有点挑战性。但是，当你批判性地看待某件事时，你不仅仅是在阅读和总结它。

你是:

- 着眼于识别 **偏见**、**空白**。
- 将它与你读过的其他研究进行比较。
- 评估 **证据的质量**。

那么，他们用来构思和撰写论文的资料是什么？你是在质疑其方法。

所以可以这样想：阅读某样东西是为了尝试弄清楚、尝试找出它的缺陷。这是一种负面的看待方式，但实际来说，就是这样。

你是在观察某样东西，以了解：

- 什么是不寻常的?
- 空白是什么?
- 比较是什么?
- 其他研究在这方面怎么说?

所以这就是 **批判性分析**。我们是在对某件事进行批判。这与总结发现是不同的。

记住，指令可能会说“批判性地分析这些论文”。明白吗?

3. 示例：总结 vs. 批判性分析

我给你们举个例子。一个非批判性的例子。然后我会展示批判性分析应该是什么样子。

Smith, 2020. 这就是我们引用论文的方式。在自主学习材料中，你们应该知道如何进行APA格式引用。**APA** 是我们在EOE中使用的引用方式。明白吗?

所以当你找到一篇论文时，有不同的分析方法。以下是一个 **非批判性总结** 的例子：

> “Smith (2020) 发现远程办公能将生产率提高15%。该研究调查了500名员工并使用了统计分析。这表明远程办公对公司是有益的。”

这是对研究的总结。这里没有任何批判性内容。

1. Absolute Fidelity Principle (Zero Information Loss - English Only)

Reem: So when you find the paper, there are different ways of doing it. Some of the questions you need to do.

Here is a **non-critical summary** example:

> "Smith (2020) found that remote work increases productivity by 15%. The study surveyed 500 employees and used statistical analysis. This shows that remote work is beneficial for companies."

This is a summary of the study. There's nothing critical here.

Now I'll show you how to write a **critical analysis**.

Example of Critical Analysis:

> "Smith's (2020) study found that remote work increases productivity by 15%. However, the study only surveyed tech workers in urban areas, which limits its generalizability. Additionally, productivity was self-reported, which may introduce bias. John's (2021) study found contrasting results using objective measures, suggesting the relationship may be more complex."

So you're saying something about the study, you're not just summarizing it. You're providing a perspective.

> 🙋‍♂️ **Speaker 2**: Find something in rebuttal.

Reem: Exactly. This is what the study is about, and this is your critical perspective about the study. This is when you look at it critically.

> 🙋‍♂️ **Speaker 2**: So I wish we could find another article, another study to make up on those--

Reem: One way, yeah. One way is to do that.

> 🙋‍♂️ **Speaker 2**: OK?

Reem: If you do that, even for comparing—even if you use one other study, it would be enough for my system. Because I know you're all starting in this journey. But do compare this with another one.

And another thing—you know how to find another one that's relevant to this one? You go to Google Scholar or whatever you want. You write the thing you're looking for. Then you'll get suggestions. You'll open the paper, you'll see the abstract. What does it say? The key points. And then you use it. You say, for example, "Another study by [Author] suggests..." but you won't just say "contrasting". OK?

So this is the **critical analysis**, and the previous one was just a **summary**. This is not an answer.

> 🙋‍♂️ **Reem**: Do you want to come to the front? 🙋‍♂️ **Speaker 2**: No, no, no. 🧑‍🏫 **Reem**: Okay, fine. You can access the slides online.

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2. Developing Research Questions

Reem: So, developing research questions. What is the gap that you're looking at in the knowledge of the literature? Always remember: **A problem well stated is a problem half solved**. If you can do the literature review well, then you're halfway through solving the research problem.

Research questions, we even call them **RQs**, should clearly define what needs to be discovered or understood.

I'll give you examples of very vague questions that will not be answerable and are not suitable for research. So, research questions should be:

- **Specific and focused**. They define the scope of the research, avoiding being too broad.
- **Significant**. Significant means there is a meaningful contribution or impact.

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5. 中文全文翻译

Reem：所以当你找到一篇论文时，有不同的分析方法。有些问题是你需要做的。

以下是一个**非批判性总结**的例子：

> "Smith (2020) 发现远程办公能将生产率提高15%。该研究调查了500名员工并使用了统计分析。这表明远程办公对公司是有益的。"

这是对研究的总结。这里没有任何批判性内容。

现在我将向你展示如何撰写**批判性分析**。

批判性分析示例：

> "Smith (2020) 的研究发现远程办公能将生产率提高15%。然而，该研究仅调查了城市地区的科技工作者，这限制了其普适性。此外，生产率是自我报告的，这可能会引入偏差。John (2021) 的研究使用客观测量得出了相反的结果，表明这种关系可能更为复杂。"

所以，你是在对研究发表看法，而不仅仅是总结它。你是在提供一个视角。

> 🙋 **Speaker 2**：找一些反驳的内容。

Reem：没错。这是研究的内容，而这是你对研究的批判性视角。这就是你批判性地审视它的时候。

> 🙋 **Speaker 2**：所以我希望我们能找到另一篇文章，另一项研究来补充那些——

Reem：一种方法，是的。这是一种方法。

> 🙋 **Speaker 2**：好的?

Reem：如果你那样做，即使是为了比较——即使你只使用另一项研究，对我的评分体系来说也足够了。因为我知道你们都刚刚开始这段旅程。但一定要将这项研究与另一项进行比较。

还有一件事——你知道如何找到另一篇与之相关的研究吗？你可以去 Google Scholar 或任何你想用的平台。输入你要找的内容。然后你会得到建议。你打开论文，会看到摘要。它说了什么？关键点。然后你就可以使用它。你可以说，例如，"另一项由 [作者] 进行的研究表明……"，但你不能只是说"相反"。明白吗？

所以这是**批判性分析**，而之前那个只是**总结**。这不是一个答案。

> 🙋 **Reem**：你想来前面吗？ 🙋 **Speaker 2**：不，不，不。 🧑‍🏫 **Reem**：好的，没关系。你可以在线访问幻灯片。

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2. 制定研究问题

Reem：那么，制定研究问题。你在文献知识中寻找的空白是什么？永远记住：**问题陈述得好，问题就解决了一半**。如果你能做好文献综述，那么你的研究问题就解决了一半。

研究问题，我们甚至称之为 **RQs**，应该明确定义需要发现或理解的内容。

我会给你举一些非常模糊的问题的例子，这些问题将无法回答，也不适合作为研究问题。因此，研究问题应该是：

- **具体且聚焦的**。它们定义了研究的范围，避免过于宽泛。
- **有意义的**。有意义意味着能做出有意义的贡献或产生影响。

1. Absolute Fidelity (English)

Reem: Research questions, we even call them **RQs**, should clearly define what needs to be discovered or understood.

I'll give you examples of very vague questions that will not be answerable and are not suitable as research questions.

So, research questions should be:

- **Specific and focused**. They define the scope of the research, avoiding being overly broad.
- **Significant**. Significant means it can make a meaningful contribution or have an impact. If I'm funding your study, there has to be enough reason for conducting that study. Why should I support you? So it has to be meaningful and have meaningful implications.

For example, if I look at the pay gaps between men and women in New Zealand. This is a topic that has been talked about a lot in the media. Women are paid less. Is this enough for me to go out and study? Yes, it is, because it is significant.

If I do this scientifically and I find out the consequences of not being paid the same as someone who is also a software engineer with me, or both of us are doctors, but he's getting more. What are the implications of that? Then I can scientifically present it to the government. They can have policies. They can reconsider the data. So it has its significance and implications. We don't do something just for the fun of it. It has to be significant.

The third, very important criterion is that the research question must be answerable.

When you come up with a study, make sure that you can answer the question. In the first week, you might still be lost about what you're going to do. Think about something that you can work on, develop, and write a proposal about in 14 weeks. Otherwise, it's not feasible.

You have to consider:

- Do I have the resources?
- Can I do this?
- Can I complete it within the time frame?

"Answerable" also means it can be investigated using research methods. And here, always remember the advice I was given: "I'm not testing your vocabulary. I'm not seeing how impressive you are. Just tell me something that I can understand easily."

So, the three criteria we use to evaluate research questions are:

1. **Significant**
2. **Focused**
3. **Feasible** (Answerable)

The question needs to have implications for the field. You need to tell me *why* answering this question is important.

You need to make sure your question is not overly broad. For example: "How do people perceive sports or physical activity?" This is too vague.

- What kind of physical activity?
- Who are you talking about?
- When? In which country? What time? What age group?

Feasible means it is realistic, considering available time, resources, and your practical strengths. This is very, very, very important.

For **transparency**, you should specify:

- **Who** the population is that you are studying.
- **What** the specific variables or phenomena are.
- **When** and **Where** the context is (e.g., in New Zealand, in China, in Ukraine).

Narrowing the scope makes the study accessible and gives key directions for data collection and analysis. The more specific, the better.

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5. 中文全文翻译

Reem：研究问题，我们甚至称之为 **RQs**，应该明确定义需要发现或理解的内容。

我会给你举一些非常模糊的问题的例子，这些问题将无法回答，也不适合作为研究问题。

因此，研究问题应该是：

- **具体且聚焦的**。它们定义了研究的范围，避免过于宽泛。
- **有意义的**。有意义意味着能做出有意义的贡献或产生影响。如果是我资助你的研究，必须有足够的理由进行这项研究。我为什么要支持你？所以它必须是有意义的，并且具有有意义的影响。

例如，如果我研究新西兰的男女薪酬差距。这是一个媒体上经常讨论的话题。女性的薪酬较低。这足以让我去研究吗？是的，因为它是有意义的。

如果我科学地进行这项研究，并找出与和我同为软件工程师的人，或者我们俩都是医生但他收入更高的人，薪酬不同的后果是什么？那会带来什么影响？然后我可以科学地向政府展示。他们可以制定政策。他们可以重新考虑数据。所以它有其重要性和影响。我们不是为了好玩而做某事。它必须是有意义的。

第三个非常重要的标准是，研究问题必须是可回答的。

当你提出一项研究时，要确保你能回答这个问题。在第一周，你可能仍然对要做什么感到迷茫。要思考一些你可以在14周内研究、发展并撰写提案的内容。否则，它就是不可行的。

你必须考虑：

- 我是否有资源？
- 我能做到吗？
- 我能在时间框架内完成吗？

"可回答"也意味着可以使用研究方法进行调查。在这里，永远记住我得到的建议："我不是在测试你的词汇量。我不是在看你的表现有多令人印象深刻。只要告诉我一些我能轻易理解的东西。"

因此，我们用来评估研究问题的三个标准是：

1. **有意义的**
2. **聚焦的**
3. **可行的**（可回答的）

这个问题需要对该领域产生影响。你需要告诉我*为什么*回答这个问题很重要。

你需要确保你的问题不会过于宽泛。例如："人们如何看待运动或体育活动？" 这太模糊了。

- 什么样的体育活动?
- 你指的是谁?
- 什么时候? 在哪个国家? 什么时间? 哪个年龄组?

可行的意味着它是现实的，要考虑到可用的时间、资源和你实际的优势。这一点非常、非常、非常重要。

为了**透明度**，你应该明确说明:

- **谁**是你正在研究的人群。
- **什么**是具体的变量或现象。
- **何时**与**何地**是研究背景（例如，在新西兰、在中国、在乌克兰）。

缩小范围使研究易于进行，并为数据收集和分析提供了关键方向。越具体越好。

3. Examples of Good Research Questions

Now, let's look at some examples of good research questions.

Example 1:

> **Does daily 30-minute aerobic exercise reduce systolic blood pressure in adults aged 40 to 60?**

This is a very good question. Let's break it down:

- **Population:** Adults aged 40 to 60.
- **Intervention:** Daily 30-minute aerobic exercise.
- **Outcome:** Reduction in systolic blood pressure.
- **Timeframe:** Daily (implied).

It is **100% clear**. I always compare formulating a research question to making a visa application. Think about all the possible things an immigration officer might ask for. If you provide all the necessary information upfront, they cannot reject your application for missing details. It's the same here. If you present this question to someone, they would not be confused.

Example 2:

> **What is the relationship between childhood food insecurity and long-term cognitive development in school-age children?**

- **Is this significant?** Yes, it is significant because it addresses a public health concern with wide-ranging implications.

Example 3:

> **How do undergraduate students at a single university perceive their satisfaction with online versus in-person learning?**

- **Is this doable?** Yes, it is feasible.
  > - I can create two groups: one using online learning and another using blended learning.
  > - I can measure and compare their performance, engagement rates, assignment submission timeliness, and their interaction patterns with teachers.
- It can realistically be completed with a survey, a limited budget, and within one semester.

Example 4:

> **Do remote workers report higher job satisfaction than those who work in an office?**

- **Is this answerable?** Yes, it is answerable. I can design a study to answer this question.

Example 5:

> **Does social media use before bedtime affect sleep quality in teenagers?**

- This question is also very clear.

Key Takeaway: For a research question to be good, it must satisfy **all** the criteria we discussed: **Focused, Researchable, Feasible, Specific, Complex, Relevant, Original, Ethical, Clear, and Significant**. Test your question against all these points.

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3. 优秀研究问题示例

现在，让我们看一些优秀研究问题的例子。

示例 1：

> 每日30分钟的有氧运动是否能降低40至60岁成年人的收缩压？

这是一个非常好的问题。我们来分解一下：

- **研究对象：** 40至60岁的成年人。
- **干预措施：** 每日30分钟的有氧运动。
- **结果：** 收缩压降低。
- **时间范围：** 每日（隐含）。

它**100%清晰**。我经常把制定研究问题比作申请签证。想想移民官可能会要求的所有可能材料。如果你提前提供了所有必要信息，他们就不能以缺少细节为由拒绝你的申请。这里也是一样。如果你向某人提出这个问题，他们不会感到困惑。

示例 2：

> 儿童期食物不安全与学龄儿童长期认知发展之间的关系是什么？

- **这具有重要性吗？** 是的，它具有重要性，因为它涉及一个具有广泛影响的公共卫生问题。

示例 3：

> 单一大学的本科生如何看待他们对在线学习与面对面学习的满意度？

- **这可行吗？** 是的，这是可行的。
    - 我可以创建两组：一组使用在线学习，另一组使用混合式学习。
    - 我可以测量并比较他们的表现、参与率、作业提交的及时性，以及他们与教师的互动模式。
- 它可以通过一项调查、有限的预算和一个学期的时间现实地完成。

示例 4：

> 远程工作者是否比在办公室工作的人报告更高的工作满意度？

- **这可以回答吗？** 是的，这是可以回答的。我可以设计一项研究来回答这个问题。

示例 5：

> 睡前使用社交媒体是否会影响青少年的睡眠质量？

- 这个问题也非常清晰。

关键要点： 一个研究问题要成为好问题，必须满足我们讨论过的**所有**标准：**聚焦的、可研究的、可行的、具体的、复杂的、相关的、原创的、合乎伦理的、清晰的、重要的。**用所有这些点来检验你的问题。

1. Absolute Fidelity Principle (Zero Information Loss - English Only)

Reem: Answerable, though it is who work and report higher job satisfaction than those who work in office. Also, I can answer them.

So answerable and feasible are basically, it's clear, the social media use before bedtime affects people related to teenagers. I think it's really clear.

Now for the question to be good, it has to be all of these. Test it.

That reminds me of something important. When you answer your assessments, I know I come back to assessments a lot. It's not the main thing when doing this. The main thing is to make use of the time. You're here, you're putting so much effort. Do learn something from it as much as you can. This is the main thing.

But again, I'm focused on assessments because this is the satisfaction you get when you do well.

So I told you a good question should cover all of these. Although in the assessment we said, significant, feasible, and focused. Okay? So not so close. Okay?

So when you answer any assessment, whether this course or any other, you'll be able to tell exactly what's the grade. Are you going to get an A plus or an A or a C minus? You can easily do that.

How? You answer according to the rubric. When we mark, we use the rubric. We don't go back to the tasks and use them again. We go like, "Okay, this is the rubric. Question 1. Did they do this? Yes, A plus." They tell you exactly. For an A plus, you do this. For an A minus, you do this.

So basically, you know when you submit your work, what your results will be. Okay?

So when you answer any assessment, I would say print out the rubric in front of you and tick off the things. Okay?

This is not relevant to software engineering because I teach this to work for MBA and LSE, but this is one way of being specific about the... This is a framework. The framework usually means a way or a method called the **PICO framework** where you, in healthcare, they tell them, okay, just in any question, make sure that you have the:

- **P**atient / **P**opulation
- **I**ntervention
- **C**omparison
- **O**utcome

A question would be, in older, the chronic back pain, this is the population. Does yoga (yoga is the intervention), compared to physiotherapy (which is the comparison), reduce the pain level (which is the outcome)?

If you can apply this to your topic, perfect. But usually it is coming from healthcare. Just to give you an idea that there are specific ways of writing good research questions. Okay?

You would be asked also to justify your research question. Whether you're writing a publication, a journal paper, a conference paper, or an assessment. So this is your research. Justify it. Show why it's important. Justify why you need to do this. So clearly explain the issue. Show the question, how the question connects to the real world.

For example, when you're...

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2. Structural Reorganization & "Breathable" Formatting (English Only)

Reem: Answerable, though it is who work and report higher job satisfaction than those who work in office. Also, I can answer them.

So answerable and feasible are basically, it's clear: the social media use before bedtime affects people related to teenagers. I think it's really clear.

Now for the question to be good, it has to be all of these. Test it.

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It's not the main thing when doing this. The main thing is to make use of the time. You're here, you're putting so much effort. Do learn something from it as much as you can. This is the main thing.

But again, I'm focused on assessments because this is the satisfaction you get when you do well.

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A question would be: In older patients with chronic back pain (this is the population), does yoga (yoga is the intervention), compared to physiotherapy (which is the comparison), reduce the pain level (which is the outcome)?

If you can apply this to your topic, perfect. But usually it is coming from healthcare. Just to give you an idea that there are specific ways of writing good research questions. Okay?

You would be asked also to justify your research question. Whether you're writing a publication, a journal paper, a conference paper, or an assessment. So this is your research. Justify it.

- Show why it's important.
- Justify why you need to do this.
- Clearly explain the issue.
- Show the question, how the question connects to the real world.

For example, when you're...

---

3. Logical Annotation & Visual Highlighting (English Only)

H3: Recap on Good Research Questions

Reem: ...Answerable, though it is who work and report higher job satisfaction than those who work in office. Also, I can answer them.

So answerable and feasible are basically, it's clear: the social media use before bedtime affects people related to teenagers. I think it's really clear.

Now for the question to be good, it has to be **all of these**. Test it.

H3: A Note on Assessments and Rubrics

That reminds me of something important. When you answer your assessments, I know I come back to assessments a lot.

It's not the main thing when doing this. The main thing is to make use of the time. You're here, you're putting so much effort. Do learn something from it as much as you can. This is the main thing.

But again, I'm focused on assessments because this is the satisfaction you get when you do well.

So I told you a good question should cover all of these. Although in the assessment we said: *significant, feasible, and focused*. Okay? So not so close. Okay?

So when you answer any assessment, whether this course or any other, you'll be able to tell exactly what's the grade. Are you going to get an A plus or an A or a C minus? You can easily do that.

How? You answer according to the **rubric**. When we mark, we use the rubric. We don't go back to the tasks and use them again. We go like, "Okay, this is the rubric. Question 1. Did they do this? Yes, A plus." They tell you exactly:

- For an A plus, you do this.
- For an A minus, you do this.

So basically, you know when you submit your work, what your results will be. Okay?

So when you answer any assessment, I would say print out the rubric in front of you and tick off the things. Okay?

H3: Frameworks for Structuring Research Questions

This is not *directly* relevant to software engineering because I teach this to work for MBA and LSE, but this is one way of being specific about the... This is a **framework**.

The framework usually means a way or a method called the **PICO framework**. In healthcare, they tell them, for any question, make sure that you have the:

- **P**atient / **P**opulation
- **I**ntervention
- **C**omparison
- **O**utcome

Example:

> In older patients with chronic back pain (this is the **population**), does yoga (yoga is the **intervention**), compared to physiotherapy (which is the **comparison**), reduce the pain level (which is the **outcome**)?

If you can apply this to your topic, perfect. But usually it is coming from healthcare. Just to give you an idea that there are specific ways of writing good research questions. Okay?

H3: Justifying Your Research Question

You would be asked also to **justify your research question**. Whether you're writing a publication, a journal paper, a conference paper, or an assessment. So this is your research. Justify it.

- Show why it's important.
- Justify why you need to do this.
- Clearly explain the issue.
- Show the question, how the question connects to the real world.

For example, when you're...

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4. Teacher-Student Interaction Extraction (English Only)

(Note: This segment of the transcript does not contain direct student questions or comments. It is a continuous lecture segment.)

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5. 中文全文翻译 (Full Chinese Translation)

H3: 关于优秀研究问题的回顾

Reem: ...是可回答的，尽管是那些在家工作的人比在办公室工作的人报告了更高的工作满意度。而且，我可以回答它们。

所以，可回答性和可行性基本上，很清楚：睡前使用社交媒体会影响与青少年相关的人。我认为这非常清楚。

现在，一个问题要成为好问题，它必须满足**所有这些**标准。用它来检验。

H3: 关于评估和评分标准的说明

这让我想起一件重要的事情。当你们回答评估时，我知道我经常回到评估这个话题。

做这件事时，评估不是主要目的。主要目的是利用好时间。你们在这里，付出了这么多努力。尽可能地从中学到东西。这才是主要目的。

但是，我再次关注评估，因为这是你们做得好时获得的满足感。

所以我告诉你们，一个好问题应该涵盖所有这些。虽然在评估中我们说的是：*重要的*、*可行的*、*聚焦的*。明白吗？所以不完全一样。明白吗？

所以，当你们回答任何评估时，无论是这门课还是其他课程，你们将能够准确判断会得到什么分数。你们是会得到A+，还是A，还是C-？你们可以很容易地做到这一点。

怎么做？ 你们根据**评分标准**来回答。我们评分时，使用评分标准。我们不会回头去看任务要求再重新使用它们。我们会这样："好的，这是评分标准。问题1。他们做了这个吗？做了，A+。" 评分标准会明确告诉你们：

- 要得到A+，你需要做到这些。
- 要得到A-，你需要做到这些。

所以基本上，当你们提交作业时，你们就知道结果会是什么。明白吗？

所以当你们回答任何评估时，我建议把评分标准打印出来放在面前，然后逐项核对。明白吗？

H3: 构建研究问题的框架

这与软件工程并非*直接*相关，因为我教这个是为了MBA和LSE的课程，但这是使问题具体化的一种方法……这是一个**框架**。

框架通常指的是一种方式或方法，称为 **PICO框架**。在医疗保健领域，他们告诉他们，对于任何问题，要确保包含：

- **P**atient / **P**opulation (患者/人群)
- **I**ntervention (干预措施)
- **C**omparison (对照)
- **O**utcome (结果)

示例：

> 在患有慢性背痛的老年患者中（这是**人群**），瑜伽（瑜伽是**干预措施**），与物理疗法（这是**对照**）相比，是否能降低疼痛水平（这是**结果**）？

如果你们能将此应用到你们的主题上，那就太好了。但这通常来自医疗保健领域。只是为了让你们了解，有一些具体的方法来撰写好的研究问题。明白吗?

H3: 论证你的研究问题

你们还会被要求**论证你的研究问题**。无论你们是在撰写出版物、期刊论文、会议论文还是评估作业。所以这是你们的研究。论证它。

- 说明它为什么重要。
- 论证你为什么需要做这个。
- 清晰地解释问题。
- 展示这个问题如何与现实世界联系起来。

例如，当你们...

H3: Justify Your Research Question

You would be asked also to **justify your research question**. Whether you're writing a publication, a journal paper, a conference paper, or an assessment. So this is your research. Justify it.

- Show why it's important.
- Justify why you need to do this.
- Clearly explain the issue.
- Show how the question connects to the real world.

For example, when you're looking at social media impacts, give an example from your homeland about what that would look like, make it grounded in reality.

- **High-position effect of authoring the research.**
- **Who will benefit from the research.**
- **What changes are contributed to the research.**

This is what justifies the importance of the question.

H3: Consequences of Poorly Formulated Research Questions

So the last thing I think we're approaching the end of the content: **consequences of formulated research questions**.

It will also be asked about commenting on why we should really focus on this. Because if it's clear, you will know how to collect the data and analyze it, which is the methodology. If it's vague, it will confuse you. So it will make it hard to select appropriate methods.

I'll give you an example.

Example 1: Vague vs. Specific

- **Vague Question:** What is the impact of online learning?
  > - This is vague, which is really important. How do I fix it?
- **Specific Question:** How does online learning affect undergraduate GPAs compared to in-person instruction?
  > - Can you see the difference? Being more specific.

It gives you unreliable findings if the question is not well defined.

So if I say, "How does education affect people?" This is so broad. Nobody can answer this. This is not scientific.

You can say, "How does completing a bachelor's degree impact entry-level salary for engineering graduates?" Specific.

Example 2: Irrelevant or Unclear Irrelevant or unusual for research, misaligningly produced findings that do not address the actual research.

If I say, "This job title relates to employee stress," there's no comparison. To make it relevant, you say, "How do workplace demands, control, and support affect employee stress levels across different job roles?"

So when you say "job," what does this job title mean? And when you say "employee stress," how? So, how do you formulate this?

> 👨‍🏫 **Reem:** Technology, Val? Why is that question? These are all two questions. What's wrong with asking this question? 🙋‍♂️ **Speaker 2:** It's not properly structured. 👨‍🏫 **Reem:** Why? 🙋‍♂️ **Speaker 2:** Let's start with giving us an idea from which range of what perspective is talking about. That's a very general question. 👨‍🏫 **Reem:** Too vague. Right? Too vague, no specific population, no specific technology, loaded language. What does "bad" mean? What does "bad" mean? So what are you talking about, and which population? So two w...

H3: 论证你的研究问题

你们还会被要求**论证你的研究问题**。无论你们是在撰写出版物、期刊论文、会议论文还是评估作业。所以这是你们的研究。论证它。

- 说明它为什么重要。
- 论证你为什么需要做这个。
- 清晰地解释问题。
- 展示这个问题如何与现实世界联系起来。

例如，当你们在研究社交媒体影响时，可以举一个你们家乡的例子来说明其具体表现，使其扎根于现实。

- **研究发表的高位效应。**
- **谁将从研究中受益。**
- **研究贡献了哪些改变。**

这就是论证问题重要性的内容。

H3: 研究问题表述不当的后果

我想我们快讲完内容了，最后一点是：**研究问题表述不当的后果。**

你们也需要评论为什么我们真的应该关注这一点。因为如果问题清晰，你们就会知道如何收集和分析数据，这就是方法论。如果问题模糊，就会让你们困惑。因此，这将使选择合适的方法变得困难。

我给你们举个例子。

示例 1：模糊 vs. 具体

- **模糊问题：** 在线学习有什么影响?
    - 这个问题很模糊，这一点非常重要。我该如何修正它?
- **具体问题：** 与面对面教学相比，在线学习如何影响本科生的平均绩点?
    - 你们能看到区别吗？要更具体。

如果问题定义不清，会导致不可靠的研究结果。

所以，如果我说“教育如何影响人们？”，这太宽泛了。没人能回答这个问题。这不科学。

你可以说“完成学士学位如何影响工程专业毕业生的起薪？”，这就具体了。

示例 2：不相关或不明确 对于研究来说，不相关或不寻常的问题会导致产生与实际问题不符的研究结果。

如果我说“这个职位头衔与员工压力有关”，这里没有比较。要使其相关，你可以说“工作场所的要求、控制和支持如何影响不同职位角色的员工压力水平？”

所以当你说“工作”时，这个职位头衔是什么意思？当你说“员工压力”时，具体指什么？那么，你该如何表述这个问题呢？

> 🧑‍🏫 **Reem:** 技术，Val？为什么是那个问题？这其实是两个问题。问这个问题有什么问题？🙋‍♂️ **Speaker 2:** 它的结构不合适。🧑‍🏫 **Reem:** 为什么？🙋‍♂️ **Speaker 2:** 首先应该让我们了解是从哪个范围、什么角度来谈论的。这是一个非常笼统的问题。🧑‍🏫 **Reem:** 太模糊了。对吧？太模糊了，没有具体的人群，没有具体的技术，使用了有倾向性的语言。“坏”是什么意思？“坏”是什么意思？所以你到底在谈论什么，以及哪个人群？所以两个 w...

1. Absolute Fidelity (English)

> 🙋‍♂️ **Speaker 2:** That's a very general question. 🧑‍🏫 **Reem:** Too vague. Right? Too vague, no specific population, no specific technology, loaded language. What does bad mean? What does bad mean? So what are you talking about, and which population?

So two ways. If you want to fix it, you can go like: how does smartphone use during study sessions affect academic performance among high school students. You see the difference. That's mature.

What about climate change? What about climate change? So yeah, it's no focus as well. So if you go like "YF people stress", too broad as well.

So you can always go through these ones and compare them to the center that you have. Make sure that you're not following these.

We're done with the test question.

We'll start with the literature review. We'll talk about the time of literature review. This is just general information coming in the middle. But we'll start with looking at the literature review, identifying the gaps.

From these gaps, we say this gap can be addressed by asking this research question.

After you answer your question, which is a long journey, you have to write your findings, right? How do we write in assessments? How do we write in reports? We write academically.

So I'll just give you what academic writing means.

So after you read and analyze, you write academically, which means a formal style of writing that follows a specific set of rules to communicate ideas clearly, precisely, and credibly.

It has to be simple. It has to be really simple.

Just recently we rejected a paper at one of the conferences because it was really important, it was really relevant and timely. And the person knew what he was doing. He was from a health agency. Very well of the language. The language is old, so bumpy. Big words.

So academic writing is not like that. Academic writing is to the point in support of my evidence.

That's why we take credit.

So if you say "climate change is causing a lot of-- oh, climate change is influencing social cohesion in New Zealand." Let's say so. Where did you get this from? This has to be signed. This has to be followed by "this 2021" or "Johnson". You have to tell me which study you get this information from.

And this is what we mean by citing.

It makes your word credible. It makes it ethical. You're not claiming that this is coming from your own research. You're giving a credit to the person who did the scientific research to come up with this. Okay?

It has a formal tone, so I think it's not acceptable. So it's not acceptable.

We have marked assessments that were really... It just gives you the idea that that person is not in the mood of the masters. So it is professional.

2. Structural Reorganization (English)

> 🙋 **Speaker 2:** That's a very general question. 🧑‍🏫 **Reem:** Too vague. Right? Too vague, no specific population, no specific technology, loaded language. What does bad mean? What does bad mean? What does bad mean? So what are you talking about, and which population?

So, there are two ways. If you want to fix it, you can phrase it like:

> "How does smartphone use during study sessions affect academic performance among high school students?"

You see the difference. That's mature.

What about "climate change"? So yeah, it's no focus as well. So if you go like "YF people stress", it's too broad as well.

So you can always go through these examples and compare them to the center that you have. Make sure that you're not following these flawed patterns.

We're done with the test question.

Transition to Literature Review

We'll start with the literature review. We'll talk about the timing of the literature review. This is just general information coming in the middle. But we'll start by looking at the literature review and identifying the gaps.

From these gaps, we say: "This gap can be addressed by asking this research question."

Introduction to Academic Writing

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It has to be simple. It has to be really simple.

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- It was really important, relevant, and timely.
- The person knew what he was doing (he was from a health agency).
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So academic writing is not like that. **Academic writing is to the point in support of my evidence.**

The Importance of Citing

That's why we take credit. So if you say:

> "Climate change is influencing social cohesion in New Zealand."

Let's say so. Where did you get this from? This has to be signed. This has to be followed by "(2021)" or "Johnson". You have to tell me which study you get this information from.

And this is what we mean by **citing**.

- It makes your word credible.
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- You're not claiming that this is coming from your own research.
- You're giving credit to the person who did the scientific research to come up with this.

Okay?

Formal Tone is Non-Negotiable

It has a formal tone, so I think it's not acceptable. So it's not acceptable.

We have marked assessments that were really... It just gives you the idea that that person is not in the mood for the masters. So it is professional.

3. Logical Highlighting (English)

H3: Refining the Research Question (Continued)

> 🙋 **Speaker 2:** That's a very general question. 🧑‍🏫 **Reem:** Too vague. Right? Too vague, no specific population, no specific technology, loaded language. What does "bad" mean? What does "bad" mean? So what are you talking about, and which population?

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> "How does smartphone use during study sessions affect academic performance among high school students?"

You see the difference. That's mature.

What about "climate change"? So yeah, it's no focus as well. So if you go like "YF people stress", it's too broad as well.

So you can always go through these examples and compare them to the center that you have. Make sure that you're not following these flawed patterns.

We're done with the test question.

H2: 2. Literature Review & Academic Writing

H3: 2.1 The Role of Literature Review

We'll start with the **literature review**. We'll talk about the timing of the literature review. This is just general information coming in the middle. But we'll start by looking at the literature review and **identifying the gaps**.

From these gaps, we say: "This gap can be addressed by asking this research question."

H3: 2.2 What is Academic Writing?

After you answer your question, which is a long journey, you have to write your findings, right? How do we write in assessments? How do we write in reports? We write **academically**.

So I'll just give you what **academic writing** means.

After you read and analyze, you write academically, which means:

- A **formal** style of writing.
- It follows a specific set of rules.
- Its purpose is to communicate ideas *clearly, precisely, and credibly*.

It has to be **simple**. It has to be really simple.

Just recently we rejected a paper at one of the conferences because:

- It was really important, relevant, and timely.
- The person knew what he was doing (he was from a health agency).
- But the language was old, so bumpy. *Big words.*

So academic writing is not like that. **Academic writing is to the point in support of my evidence.**

H3: 2.3 The Critical Practice of Citing

That's why we **take credit**. So if you say:

> "Climate change is influencing social cohesion in New Zealand."

Let's say so. *Where did you get this from?* This has to be signed. This has to be followed by "(2021)" or "Johnson". You have to tell me which study you get this information from.

And this is what we mean by **citing**.

- It makes your word **credible**.
- It makes it **ethical**.
- You're not claiming that this is coming from your own research.
- You're giving credit to the person who did the scientific research to come up with this.

Okay?

H3: 2.4 The Requirement of a Formal Tone

It has a **formal tone**, so I think it's not acceptable. So it's not acceptable.

We have marked assessments that were really... It just gives you the idea that that person is *not in the mood of the masters*. So it is **professional**.

4. Interaction Extraction (English)

> 🙋 **Speaker 2:** That's a very general question. 🧑‍🏫 **Reem:** Too vague. Right? Too vague, no specific population, no specific technology, loaded language. What does bad mean? What does bad mean? So what are you talking about, and which population?

So two ways. If you want to fix it, you can go like: how does smartphone use during study sessions affect academic performance among high school students. You see the difference. That's mature.

What about climate change? What about climate change? So yeah, it's no focus as well. So if you go like "YF people stress", too broad as well.

So you can always go through these ones and compare them to the center that you have. Make sure that you're not following these.

We're done with the test question.

We'll start with the literature review. We'll talk about the time of literature review. This is just general information coming in the middle. But we'll start with looking at the literature review, identifying the gaps.

From these gaps, we say this gap can be addressed by asking this research question.

After you answer your question, which is a long journey, you have to write your findings, right? How do we write in assessments? How do we write in reports? We write academically.

So I'll just give you what academic writing means.

So after you read and analyze, you write academically, which means a formal style of writing that follows a specific set of rules to communicate ideas clearly, precisely, and credibly.

It has to be simple. It has to be really simple.

Just recently we rejected a paper at one of the conferences because it was really important, it was really relevant and timely. And the person knew what he was doing. He was from a health agency. Very well of the language. The language is old, so bumpy. Big words.

So academic writing is not like that. Academic writing is to the point in support of my evidence.

That's why we take credit.

So if you say "climate change is causing a lot of-- oh, climate change is influencing social cohesion in New Zealand." Let's say so. Where did you get this from? This has to be signed. This has to be followed by "this 2021" or "Johnson". You have to tell me which study you get this information from.

And this is what we mean by citing.

It makes your word credible. It makes it ethical. You're not claiming that this is coming from your own research. You're giving a credit to the person who did the scientific research to come up with this. Okay?

It has a formal tone, so I think it's not acceptable. So it's not acceptable.

We have marked assessments that were really... It just gives you the idea that that person is not in the mood of the masters. So it is professional.

5. Full Chinese Translation

H3: 精炼研究问题（续）

> 🙋 **发言者 2:** 这是一个非常笼统的问题。 🧑‍🏫 **Reem:** 太模糊了。对吧？太模糊了，没有具体的人群，没有具体的技术，使用了有倾向性的语言。“坏”是什么意思？“坏”是什么意思？所以你到底在谈论什么，以及哪个人群？

所以，有两种方式。如果你想修正它，你可以这样表述：

> “在学习时段使用智能手机如何影响高中生的学业表现？”

你看出区别了吗？这样才是成熟的。

那“气候变化”呢？所以，是的，它同样没有焦点。所以如果你说“YF 人群压力”，也同样太宽泛了。

所以你可以随时通过这些例子，与你心中的核心标准进行比较。确保你没有遵循这些有缺陷的模式。

我们关于测试问题就讲到这里。

H2: 2. 文献综述与学术写作

H3: 2.1 文献综述的作用

我们将从**文献综述**开始。我们会讨论进行文献综述的时机。这只是中间插入的一般信息。但我们将从审视文献综述和**识别研究空白**开始。

从这些空白出发，我们说："这个空白可以通过提出这个研究问题来解决。"

H3: 2.2 什么是学术写作?

在你回答了你的问题之后——这是一个漫长的过程——你必须写下你的发现，对吧？我们如何在评估中写作？我们如何在报告中写作？我们进行**学术写作**。

所以，我来告诉你们**学术写作**意味着什么。

在你阅读和分析之后，你进行学术写作，这意味着：

- 一种**正式的**写作风格。
- 它遵循一套特定的规则。
- 其目的是*清晰*、*精确*、*可信地*传达思想。

它必须是**简洁的**。必须非常简洁。

就在最近，我们拒绝了一篇会议论文，因为：

- 它非常重要、相关且及时。
- 作者知道自己在做什么（他来自一个卫生机构）。
- 但语言陈旧，非常拗口。*用了很多大词。*

所以学术写作不是那样的。**学术写作是切中要害，以支持我的证据。**

H3: 2.3 引用的关键实践

这就是为什么我们要**注明出处**。所以如果你说：

> "气候变化正在影响新西兰的社会凝聚力。"

假设是这样。*你从哪里得到这个信息的？* 这必须署名。这后面必须跟上"(2021年)"或"Johnson"。你必须告诉我你是从哪项研究中获得这个信息的。

这就是我们所说的**引用**。

- 它使你的话**可信**。
- 它使其**合乎道德**。
- 你不是在声称这来自你自己的研究。

- 你是在向进行这项科学研究并得出此结论的人**致谢**。

明白吗?

H3: 2.4 正式语态的要求

它必须具有**正式的语态**，所以我认为非正式语态是不可接受的。所以这是不可接受的。

我们批改过一些作业，真的……这只会让你觉得那个人*没有进入硕士学习的状态*。所以它必须是**专业的**。

H3: 2.5 学术写作的语言特征

它必须是**专业的语言**，而不是随意或对话式的表达。

它是经过编辑的，并且是基于研究的，或者说基于数据的。

其核心是关于研究，而非个人观点。

它拥有清晰的结构，因此在逻辑上是严密的。

并且它使用精确的语言。

我们不使用模糊的词汇，而是使用**特定的术语**。

我们不说“我认为”或“我做了那个”或“我收集了数据然后我分析了它”。

没有“我”、“我们”或“你”，要用**第三人称**书写。

例如：“数据已被收集，数据已被分析”。

明白吗?

这些只是供你参考的软件工程领域的例子。

H4: 非正式语言的例子

例如，如果说：

> “网络安全不仅仅是关于防火墙。公司需要认真保护他们的东西。”

“东西”这个词太随意了，对吧？

一个好的表述应该是：

> “网络安全超越了防火墙的实施，涵盖了全面的威胁评估、员工培训和事件响应协议。”

你看出区别了吗？

> 🙋‍♂️ Student: 如何判断我的写作是否科学？ 👨‍🏫 Lecturer: 一种方法是使用AI。你可以问：“这篇写作有什么问题？ ”

请记住，我们说过我们并非禁止使用AI，而是应该**有效地使用它**。

例如，用于检查、提供示例、组织语言、校对语法等，这是完全可以的。

但是，如果我们觉得这份作业不真实，或者这个人只是从ChatGPT或其他工具上复制粘贴，那么真正的问题就来了。

H4: 无依据的主张

这一点也很重要。

例如：

> “每个人都知道社交媒体对青少年有害。它显然会导致抑郁和焦虑。”

像“显然”、“每个人都知道”这样的表述是不严谨的。

事实上，我们从不认为有什么事情是100%正确的。即使它是，我们也总是尝试为质疑留出一些空间，为他人与你讨论留出余地。

所以，避免使用“每个人都知道”、“显然”这类词，避免在没有提供实际证据的情况下做出笼统的概括。

这些都是**缺乏支持**的表述。

> 🙋‍♂️ Student: 我如何让它可信？ 👨‍🏫 Lecturer: 我使用**资源**。

例如：

> “最近的实证研究（Smith & Arthur, 2023）已经确定了过度使用社交媒体与抑郁发生率增加之间的相关性。”

这样，我说明了信息的来源（Smith和Arthur），并且告诉了你它有多新。

同时，也可以补充说明：

> “尽管研究人员仍在争论这种关系是否是因果关系。”

H3: 2.5 学术写作的语言特征

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我们不使用模糊的词汇，而是使用**特定的术语**。

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同时，也可以补充说明：

> “尽管研究人员仍在争论这种关系是否是因果关系。”

Session Summary and Practical Exercise

Summary of Session Content

I don't want you to get too confused with all the information we've covered.

I was speaking to put you in the picture of research and academic writing, such as:

- The types of literature reviews we have.
- What we consider credible.

But the main content of this session, in a nutshell, is:

1. **Conduct research.**
2. **Conduct a literature review.**
3. **Identify the gap.**

For each gap you identify, you formulate a **research question**.

You must ensure that this research question is:

- **Feasible**
- **Specific**
- **Significant**

That's the core of it.

> 🧑‍🏫 **Lecturer:** So, does anyone have any questions? Any questions?

🙋‍♂️ **Student:** The last slide said... just... So we will be beyond that...

🧑‍🏫 **Lecturer:** You need to be very clear. Why? If I say "studies show," and if I say "one study shows," it's completely different.

We need to be very specific. The writing has to be polished. It has to be up to the standards of being published in a reputable journal.

So yes, you're right. For example, things like that—yes, you need to upgrade the level of language you're using.

And that's why I said there is an **ethical use of AI**, because English is not your first language.

But still, you need to go back and read papers to understand the knowledge.

Introduction to the Practical Exercise

Now, I don't think we will have time to cover the whole assessment—sorry, the exercise—but it's very important to go through it.

For those of you who are often on the Blackboard, go to the exercise in the self-directed learning section. I'll try to go through it here.

I think the most challenging part of the assessment is this part, not developing the research question itself. I think you'd be able to formulate the question quite easily.

The most challenging part is to be able to read the paper and identify the gaps. From the gaps, you can easily formulate the question.

Exercise Instructions

In this exercise, you will read two short academic-style literature snippets drawn from the fields of **process informatics** and **information systems**.

Procedure:

1. Before discussing the snippets, read each one individually first.
2. Then, discuss it with your peers to ensure everyone has a shared understanding.

The two sections I took are fabricated. There is no actual paper that says that. It's just something I made up; even the references, some of them are not real. These are just for practice.

Your Task: I want you to read and discuss the snippets. I want you to:

1. Identify the gap.
2. Formulate the research questions.

This exercise will be in three parts.

课程总结与实操练习

课程内容总结

我不希望你们被我们讨论的所有信息搞得太困惑。

我讲解的目的是让你们了解研究和学术写作的概况，例如：

- 我们有哪些类型的文献综述。
- 我们认为什么是可信的。

但本次课程的核心内容，简而言之是：

1. **进行研究。**
2. **进行文献综述。**
3. **识别研究空白。**

对于你识别的每一个研究空白，你需要提出一个**研究问题**。

你必须确保这个研究问题是：

- **可行的**
- **具体的**
- **有意义的**

这就是核心。

🧑‍🏫 **讲师：** 那么，有人有问题吗？有任何问题吗？

🙋 **学生：** 最后一张幻灯片说……只是……所以我们将会超越那个……

🧑‍🏫 **讲师：** 你需要非常清晰。为什么？如果我说“研究表明”，和我说“一项研究表明”，是完全不同的。

我们需要非常具体。写作必须经过润色。它必须达到能在知名期刊上发表的标准。

所以，是的，你是对的。例如，像那样的事情——是的，你需要提升你所使用的语言水平。

这就是为什么我说存在**人工智能的伦理使用**，因为英语不是你们的母语。

但你们仍然需要回去阅读论文，以理解其中的知识。

实操练习介绍

现在，我认为我们没有时间涵盖整个评估——抱歉，是整个练习——但完成它非常重要。

对于经常使用Blackboard的同学，请前往自主学习部分的练习。我会在这里尝试讲解一下。

我认为评估中最具挑战性的部分是这一部分，而不是制定研究问题本身。我认为你们应该能够相当轻松地提出问题。

最具挑战性的部分是能够阅读论文并识别出研究空白。 从研究空白出发，你可以轻松地提出问题。

练习说明

在这个练习中，你将阅读两段简短的学术风格文献摘录，内容来自**流程信息学**和**信息系统**领域。

步骤：

1. 在讨论摘录之前，首先单独阅读每一段。
2. 然后，与你的同伴讨论，以确保每个人都有共同的理解。

我选取的这两段内容是虚构的。并没有实际存在的论文这么说。这只是我编造的东西；甚至参考文献，有些也不是真实的。这些仅供练习使用。

你们的任务： 我希望你们阅读并讨论这些摘录。我希望你们：

1. 识别研究空白。
2. 制定研究问题。

这个练习将分为三个部分。

Reem: It's just something that I made, even the references, some of them are not. These are just done.

I want you to read and discuss the snippets. I think we might have some.

I want you to read and discuss the snippets. I want you to identify the gap and then do the research questions. Okay?

So this will be three parts.

Part One:

- Read the snippets.
- Underline or highlight any claim that appears strongly supported by psychic evidence.

🙋‍♂️ **Student**: If you don't understand the question, stop me. 🧑‍🏫 **Reem**: Okay?

Part Two: So what you're going to do is:

1. Individually read the snippets.
2. Then turn to the person sitting next to you.
    - Even if you want to do it in your own language, that's not a point.
    - Make sure that this is what I understood. What did you understand from this piece?
    - Ensure that you're both on the same page.

Okay.

Part Three: So the one in blue and the one in yellow. These are references.

Remember when I told you any scientific piece has to support its advice with something credible? Then there will be different reasons. When I evaluate the paper, I'll look at these ones. If they're good, then you'll find that the paper is good as well. Okay?

So the first part is easy. Read it and make sure that you understand it.

Identifying the research gap is the standard one. How do I identify a gap?

Okay. Considering the truth of the snippets, identify at least two explicit gaps. And one explicit gap. It's too complicated for you at this stage to identify the implicit gaps. I want you just to look at the snippets and identify **explicit gaps**.

How does that work? Look for claims that, for example, have multiple citations or no **hedging language**.

Hedging language means like, "this is happening, however..." or these kinds of things, not "but". "However", "nevertheless", these are the words we use instead of using "but".

So just go through this, try to work around it by identifying gaps. That means things that are not covered in the study that you can then do on your own in your new study.

> 🙋 **Student**: Does that make sense? 👩‍🏫 **Reem**: I want you to ask, is it a little bit complicated?

Just read the two letters, the blue and yellow. This person is the person next to you, identify that in this study. And then we will do it.

Thank you.

You have done with this one. If you finish this one and you discuss it with your colleagues, you're ready for that assessment. I can guarantee that you'll get good marks, okay?

But just because it might be a little bit challenging for you, I'll just go through how we identify gaps.

Forget about **implicit ones**. Implicit ones requires a lot of reading, a lot of research when you're conducting your own.

Focus on the **explicit research gaps**.

So if you look at evaluating the evidence, if it's supported by more than one reference, the more references that you use to support, the better and the more credible your statement is.

If you find thi...

【中文全文翻译】

Reem: 这只是我编造的东西，甚至参考文献，有些也不是真实的。这些仅供练习使用。

我希望你们阅读并讨论这些摘录。我想我们可能会有一些。

我希望你们阅读并讨论这些摘录。我希望你们识别研究空白，然后制定研究问题。明白吗?

所以这将分为三个部分。

第一部分：

- 阅读摘录。
- 在那些看起来有强有力的心理证据支持的论断下划线或高亮。

> 🙋‍♂️ **学生**：如果你不理解这个问题，请打断我。 👩‍🏫 **Reem**：好的?

第二部分： 所以你们要做的是：

1. 单独阅读每一段摘录。
2. 然后转向坐在你旁边的人。
    - 即使你想用自己的语言讨论，这也没关系。
    - 确保你的理解是这样的。你从这段内容中理解到了什么?
    - 确保你们双方有共同的理解。

好的。

第三部分： 所以蓝色和黄色的这两段。这些是参考文献。

还记得我告诉过你们，任何科学文章都必须用可靠的东西来支持其论点吗?然后会有不同的理由。当我评估论文时，我会看这些参考文献。如果它们质量好，那么你会发现论文本身也很好。明白吗?

所以第一部分很简单。阅读并确保你理解了。

识别研究空白是标准步骤。我如何识别一个空白?

好的。考虑到这些摘录的真实性，识别至少两个显性空白。以及一个显性空白。在这个阶段，识别隐性空白对你们来说太复杂了。我只希望你们看着这些摘录，识别**显性空白**。

这怎么操作？寻找那些，例如，有多个引用或没有**模糊限制语**的论断。

模糊限制语指的是像“这正在发生，然而……”这类表达，而不是“但是”。“然而”、“尽管如此”，这些是我们用来替代“但是”的词。

所以，通读这些内容，尝试通过识别空白来解决问题。空白指的是在该研究中未涵盖的、你可以在自己的新研究中去做的事情。

> 🙋 **学生**：这说得通吗？ 👩‍🏫 **Reem**：我希望你们问一下，这有点复杂吗？

只需阅读这两封信，蓝色和黄色的。这个人就是你旁边的人，在这项研究中识别出来。然后我们就开始做。

谢谢。

你们已经完成了这个。如果你们完成了这个并与同事讨论过，你们就为那个评估做好了准备。我可以保证你们会得到好成绩，好吗？

但正因为这对你们来说可能有点挑战性，我将简要介绍一下我们如何识别空白。

忘掉**隐性空白**。隐性空白需要大量的阅读和研究，当你们自己进行研究时。

专注于**显性研究空白**。

所以，如果你看评估证据，如果它被不止一个参考文献支持，你用来支持的参考文献越多，你的陈述就越好、越可信。

如果你发现这...

3.2.2 Identifying Explicit Research Gaps

Reem: Focus on the **explicit research gaps**.

So if you look at evaluating the evidence:

- If it's supported by more than one reference, the more references you use to support it, the better and the more credible your statement is.
- If you find things that are written without any support or any references, this is weak support. These would be assumptions by the authors themselves.

Example from Snippet A

Pay attention to snippet A. An explicit gap would be:

> "There is limited published guidance on how to design decision support systems that function reliably when data quality is poor. The authors note that existing evaluations rely on clean, pre-processed data, creating a gap between lab platforms and real-world conclusions."

So anything where the author is saying "this is happening, however, this is not happening" or "this is not well supported" indicates a gap. Maybe you can test whether they're right or wrong.

Example from Snippet B

Let me give you another example. An explicit gap from snippet B would be:

> "The evidence base is concentrated in well-resourced, high-income country hospital settings. Whether findings cover low-resource settings, community health centers, or field hospitals in disaster zones is not known."

You will see that they're talking about well-resourced hospitals. What about other types of hospitals?

So, when we formulate the research questions corresponding to the gap, we can say that studies do not consider or are only focused on well-resourced hospitals. You can think about low-income countries, low-resource hospitals, and so on.

3.2.3 Student Interaction: Identifying Gaps

Reem: Don't worry about that. I'll just post the answer, but if anyone could identify a gap, please share it with us. Was anyone able to identify a gap from A? Or is the language too complicated?

> 🙋‍♂️ **Speaker 2**: "It therefore remains unclear whether findings are translated..." I think it is also explicit.
>
> 🧑‍🏫 **Reem**: Correct. Yeah. The first part.
>
> 🙋‍♂️ **Speaker 2**: It doesn't say like how it was before? It only says "it remains unclear"...

> 👨‍🏫 **Reem**: Yes. So any idea or anything where you think something was covered and something else was excluded, it's an invitation for us to look for or to research the excluded aspect.

Reem: Anyone can figure out something else?

> 🙋‍♂️ **Speaker 2**: I think they were talking about hospital systems in high-income countries. What about other countries?
>
> 👨‍🏫 **Reem**: Correct. Yeah.

Reem: Okay. So anything that gives you information, think about that. What's the context of the research? You can also...

3.2.2 识别显性研究空白

Reem: 专注于**显性研究空白**。

因此，在评估证据时：

- 如果它被不止一个参考文献支持，你用来支持的参考文献越多，你的陈述就越好、越可信。
- 如果你发现一些内容是在没有任何支持或任何参考文献的情况下写成的，这就是弱支持。这些将是作者自己的假设。

来自片段 A 的示例

注意片段 A。一个显性空白会是：

> "关于如何设计在数据质量差时仍能可靠运行的决策支持系统，已发表的指导有限。作者指出，现有的评估依赖于干净、预处理过的数据，这在实验室平台和现实世界结论之间造成了差距。"

所以，任何作者说"这种情况正在发生，然而，那种情况没有发生"或"这没有得到很好的支持"的地方，都表明存在空白。也许你可以测试他们是对还是错。

来自片段 B 的示例

让我再举一个例子。来自片段 B 的一个显性空白会是：

> "证据基础集中在资源充足的高收入国家医院环境中。研究结果是否涵盖资源匮乏的环境、社区卫生中心或灾区的野战医院尚不清楚。"

你会看到他们谈论的是资源充足的医院。那么其他类型的医院呢？

因此，当我们制定与空白相对应的研究问题时，我们可以说研究没有考虑或只关注资源充足的医院。你可以考虑低收入国家、资源匮乏的医院等等。

3.2.3 师生互动：识别空白

Reem: 别担心那个。我会直接公布答案，但如果有人能识别出一个空白，请与我们分享。有人能从 A 中识别出一个空白吗？还是语言太复杂了？

> 🙋 **学生 2**: "因此，研究结果是否被转化仍不清楚……" 我认为这也是显性的。
>
> 🧑‍🏫 **Reem**: 正确。是的。第一部分。
>
> 🙋 **学生 2**: 它没有说以前是怎样的？它只说"仍不清楚"……
>
> 🧑‍🏫 **Reem**: 是的。所以，任何你觉得某些方面被涵盖而另一些方面被排除的想法或事情，都是邀请我们去寻找或研究被排除的方面。

Reem: 有人能想出别的吗？

> 🙋 **学生 2**: 我认为他们谈论的是高收入国家的医院系统。那么其他国家呢？
>
> 🧑‍🏫 **Reem**: 正确。是的。

Reem: 好的。所以，任何给你信息的东西，都思考一下。研究的背景是什么？你也可以...

1. Absolute Faithfulness (English)

Reem: Anyone can figure out something else?

🙋 **Speaker 2**: I think they were talking about hospital systems and high income countries. What about other countries?

🧑‍🏫 **Reem**: Correct. Yeah.

Reem: Okay. So anything that would give you an information, think about that. What's the context of the research?

You can also think about: we have conducted a qualitative study. Remember last time we just gave a slide idea about what qualitative and quantitative means.

How about I do the same thing, complicated, and see whether they're trying to do it or not. Trying to do it means, do they give me the same answer? If not, this is even a potential for research.

So maybe I can choose a different methodology. I can choose a different setting. I can choose something within the same setting, same methodology, but another aspect. Okay?

The thing here that I wanted to make is the thing that I feel well supported. This you would use when you're writing your own assessment.

So if you want me to really think or to trust your body, cite it. The more citations, the more credible. Okay?

Clips that appear weak or unsupported, single citations, although that's not always the case. So in practice, in reality, if it's a well-supported, if it's a, for example, something supported by the C.E.O. or IEEE, we will take it as credible. Okay?

Assumptions or no citation at all. No citation could be assumptions by the authors, which is an invitation for you to make sure that it's right or wrong.

Pay attention to **hedging language**, which is used to make statements less absolute or less direct, especially in academic writing.

Things like, if I say: "This causes errors in the system." This is a very strong thing, right? Compare it to this mild language: "This may cause errors in the system." This is the **hedging language**. It may, it means that it may not. So I can develop a question to test whether it causes errors in the system. Okay?

Another thing would be: what is missing from this? Identifying if this is a **gap**.

As I told you, we're not focused so much on the emphasis at this stage, but you can, for example, think about the groups that they were talking about. So things like-- we did not specifically say however, or nevertheless, or. But you can think about it from the context.

So are any groups absent? Are geographic or socioeconomic groups not represented? For example, you've done a study in New Zealand. How about if I do it in China and compare?

Technologies, I don't know where the technology is not discussed. So if I'm talking about revolution in technology, and then I talk about everything and not AI, so maybe I can focus on AI.

So all of these are places or areas where you can identify opportunities.

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5. 中文全文翻译

Reem: 有人能想出别的吗?

> 🙋‍♂️ **发言者 2**: 我认为他们谈论的是高收入国家的医院系统。那么其他国家呢?
>
> 🧑‍🏫 **Reem**: 正确。是的。

Reem: 好的。所以，任何能给你信息的东西，都思考一下。研究的背景是什么?

你也可以思考：我们进行了一项定性研究。记得上次我们只是用一张幻灯片介绍了定性和定量的含义。

如果我做同样复杂的事情，看看他们是否试图去做，会怎么样？试图去做意味着，他们是否给了我相同的答案？如果没有，这甚至是一个潜在的研究机会。

所以，也许我可以选择不同的方法论。我可以选择不同的环境。我可以在相同的环境、相同的方法论下，选择另一个方面。明白吗?

这里我想强调的是，我觉得有充分支持的东西。这在你们撰写自己的评估报告时会用到。

所以，如果你真的想让我思考或相信你的论述，请引用它。引用越多，可信度越高。明白吗?

那些显得薄弱或缺乏支持的片段，单一的引用，尽管情况并非总是如此。所以，在实践中，在现实中，如果它得到了充分的支持，例如，得到了 C.E.O. 或 IEEE 支持的东西，我们会认为它是可信的。明白吗?

假设或根本没有引用。没有引用可能是作者的假设，这邀请你去验证它是对是错。

注意**模糊限制语**，它用于使陈述不那么绝对或直接，尤其是在学术写作中。

例如，如果我说："这会导致系统中的错误。"这是一个非常强烈的说法，对吧？把它与这种温和的语言比较："这可能导致系统中的错误。"这就是**模糊限制语**。"可能"，意味着它也可能不会。所以我可以设计一个问题来测试它是否会导致系统中的错误。明白吗?

另一件事是：这里面缺少了什么？识别这是否是一个**研究空白**。

正如我告诉你们的，在这个阶段我们不太关注强调，但你们可以，例如，思考他们谈论的群体。所以，像——我们没有明确说"然而"、"不过"或"但是"。但你们可以从上下文中思考。

那么，是否有任何群体缺席？是否有地理或社会经济群体未被代表？例如，你在新西兰做了一项研究。如果我在中国做并比较呢?

技术，我不知道哪里没有讨论技术。所以，如果我在谈论技术革命，然后我谈论了所有东西却没有提到人工智能，那么也许我可以专注于人工智能。

所以，所有这些都是你可以识别机会的地方或领域。

Identifying Research Opportunities: Data Types and Temporal Scope

Reem: For example, you've done a study in New Zealand.

> 🙋 **Student**: How about if I do it in China and compare?

Reem: Technologies, I don't know where the technology is not discussed. So if I'm talking about a revolution in technology, and I talk about everything *except* AI, then maybe I can focus on AI.

So, all of these are places or areas where you can identify opportunities for further research.

- **The data types.**
    - Are you, for example, collecting interviews? I could do it for today's data.
    - Are you collecting audio?
    - Are you using videos?
    - There are so many ways. We'll see that in data collection methods.
- **The temporal scope.**

> - If I'm looking at the response to earthquakes, am I looking at the **immediate response**, or the long-term welfare impact, like three years down the line? These kinds of temporal distinctions are crucial.

So, basically, that's it.

Reem: Do go through the exercise, and I'll post the model answers the following day. This should help clarify any questions.

Speaker 2: So, thank you, and see you next week. 太好了 (Excellent).

Summary & Resources

1. Administrative Notices Summary

- The lecturer will post model answers for the discussed exercise on the following day.
- The next session is scheduled for the coming week.

2. Core Academic Knowledge Points

- **Identifying Research Gaps through Comparison:** Opportunities can arise from applying a study's context (e.g., geographical location like China vs. New Zealand) or focus (e.g., a specific technology like AI within a broader tech revolution) to new settings.
- **Methodological Expansion as Research Opportunity:** Novel research can be designed by varying the **data types** (e.g., interviews, audio, video) and **data collection methods** used in existing studies.
- **Temporal Scope as a Research Dimension:** The timeframe of investigation (e.g., immediate vs. long-term effects in disaster response studies) presents a significant axis for developing new research questions.

3. Recommended Learning Resources (Level 9)

- **Textbook:** Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (5th ed.). SAGE Publications. (Chapters on formulating research problems and questions are highly relevant).
- **Academic Paper:** Alvesson, M., & Sandberg, J. (2011). Generating research questions through problematization. *Academy of Management Review, 36*(2), 247-271. This paper

provides a sophisticated framework for identifying research opportunities by challenging assumptions in existing literature.
- **Methodology Guide:** The "SAGE Research Methods" online platform. It offers advanced content on research design, including dedicated modules on gap identification and question formulation suitable for postgraduate research.

中文全文翻译

识别研究机会：数据类型与时间范围

Reem: 例如，你在新西兰做了一项研究。

> 🙋 **学生**：如果我在中国做并比较呢？

Reem: 技术，我不知道哪里没有讨论技术。所以，如果我在谈论技术革命，然后我谈论了所有东西却没有提到人工智能，那么也许我可以专注于人工智能。

所以，所有这些都是你可以识别进一步研究机会的地方或领域。

- **数据类型。**
  - 例如，你是在收集访谈吗？我可以针对当今的数据来做。
  - 你是在收集音频吗？
  - 你是在使用视频吗？
  - 方法有很多。我们将在数据收集方法中看到这一点。
- **时间范围。**
  - 如果我在研究对地震的响应，我是在看**即时响应**，还是长期福利影响，比如三年后的情况？这类时间上的区分至关重要。

所以，基本上就是这样。

Reem: 请务必完成这个练习，我将在第二天发布参考答案。这应该有助于澄清任何问题。

Speaker 2: 那么，谢谢大家，下周见。太好了。

总结与资源推荐

1. 事务性通知汇总

- 讲师将在次日发布所讨论练习的参考答案。
- 下一次课程安排在下一周。

2. 核心学术知识点提炼

- **通过比较识别研究空白：** 将一项研究的背景（例如，中国与新西兰的地理位置）或焦点（例如，在更广泛的技术革命中专注于人工智能等特定技术）应用于新的环境，可以产生研究机会。
- **方法论扩展作为研究机会：** 通过改变现有研究中使用的**数据类型**（例如，访谈、音频、视频）和**数据收集方法**，可以设计出新颖的研究。
- **时间范围作为一个研究维度：** 调查的时间框架（例如，灾害响应研究中的即时效应与长期效应）是提出新研究问题的一个重要轴线。

3. 推荐学习资源（9级水平）

- **教科书：** Creswell, J. W., & Creswell, J. D. (2018). *研究设计：定性、定量与混合方法路径* (第5版). SAGE Publications. （其中关于构建研究问题和提出研究假设的章节高度相关）。
- **学术论文：** Alvesson, M., & Sandberg, J. (2011). 通过问题化生成研究问题. *Academy of Management Review, 36*(2), 247-271. 这篇论文提供了一个复杂的框架，通过挑战现有文献中的假设来识别研究机会。
- **方法论指南：** "SAGE Research Methods" 在线平台。它提供关于研究设计的高级内容，包括适合研究生研究的、专门针对空白识别和问题构建的模块。